ABSTRACT

African American women experience approximately 40% higher breast cancer mortality than non-Hispanic White women despite similar incidence rates, a disparity driven by incompletely characterized biological and environmental factors. Arsenic, an established environmental carcinogen with evidence for racial variation in exposure, is a known inhibitor of nucleotide excision repair (NER), but whether this inhibition differs by racial background or is amplified in the breast cancer stem cell compartment has not been investigated. Here we report transcriptome-wide profiling of two breast tissue-derived cell lines representing African American (BRL-24) and non-Hispanic White (BRL-23) racial backgrounds across parental and cancer stem cell-enriched populations under vehicle and chronic low-dose arsenic trioxide conditions (0.3 μM, 14 days, approximately the LC10). Cancer stem cells were isolated by flow cytometric sorting using CD24, CD49f, and CD44 surface co-expression (CD24−/CD49f+/CD44+), established markers of breast tissue-derived progenitor populations.[6,7] RNA from all 16 samples was profiled by paired-end RNA sequencing and analyzed using DESeq2 with a full factorial interaction model followed by GSEAPreranked across four MSigDB databases (48 total runs; significance: |NES| > 1.5, FDR < 0.25). Sorted populations were validated as genuine cancer stem cells by enrichment of published embryonic and mammary stem cell gene signatures and coordinate elevation of BER, MMR, and HR pathway programs. While arsenic collapsed 68% of baseline racial transcriptional differences in parental cells, racial differences were amplified approximately tenfold in the stem cell compartment (582 versus 71 significantly enriched pathways). The primary finding was selective suppression of *KEGG_NUCLEOTIDE_EXCISION_REPAIR* specifically in African American stem cells under arsenic (NES = −2.75, FDR = 0.18), confirmed independently by Reactome (NES = −2.68), affecting both the damage recognition stage (XPC, DDB1, CETN2, RAD23A, ERCC3, GTF2H1) and the gap-filling and ligation machinery (PCNA, LIG1, RFC3–5). No equivalent suppression was observed in non-Hispanic White stem cells under arsenic (NES = +0.86) nor in parental cells of either line under any condition, establishing that the racial NER disparity requires the simultaneous combination of African American cellular biology, cancer stem cell identity, and arsenic exposure. *ERCC1*, the obligate NER incision endonuclease subunit, was consistently suppressed in African American stem cells across four comparisons and present in both independent racial NER leading edges, identifying it as the primary candidate for functional validation. These findings provide the first transcriptomics-level mechanistic framework connecting racial background, arsenic exposure, cancer stem cell biology, and impaired genomic maintenance relevant to breast cancer health disparities.

Keywords: *breast cancer; racial disparities; arsenic trioxide; nucleotide excision repair; ERCC1; cancer stem cells; CD24−/CD49f+/CD44+; RNA sequencing; gene set enrichment analysis; health disparities*

INTRODUCTION

Breast cancer is the most frequently diagnosed cancer and the second leading cause of cancer death among women in the United States.[27] Significant racial disparities persist in breast cancer outcomes: African American (AA) women experience approximately 40% higher breast cancer mortality than non-Hispanic White (NHW) women and are more likely to be diagnosed with aggressive, high-grade tumors at younger ages.[28,29] Environmental exposures including endocrine-disrupting chemicals and heavy metals are increasingly recognized as contributors to these disparities, yet the specific mechanisms by which racial background modifies vulnerability to environmental carcinogens remain incompletely characterized.[30]

Arsenic is a Group 1 human carcinogen classified by the International Agency for Research on Cancer and an established environmental carcinogen with documented associations with breast cancer risk, particularly for premenopausal women and with evidence for racial variation in tissue-level sensitivity.[31] In the United States, arsenic contamination of drinking water affects millions of people, with particularly high levels documented in communities across South Florida and other Southern states that also carry elevated rates of advanced cancer diagnosis.[4] Arsenic promotes carcinogenesis through multiple mechanisms including oxidative stress, epigenetic dysregulation, and - critically - inhibition of DNA repair.[32,33] Unlike classical mutagens that directly cause point mutations, arsenic acts primarily as a co-carcinogen by impairing the capacity of cells to remove DNA damage caused by other environmental insults, amplifying their mutagenic effect.[34]

Nucleotide excision repair (NER) is the primary DNA repair pathway responsible for removing a broad spectrum of helix-distorting lesions, including bulky adducts induced by polycyclic aromatic hydrocarbons, UV radiation, and arsenic-associated oxidative damage.[35,36] NER operates through two sub-pathways: global-genome NER (GG-NER), which surveys the entire genome for lesions, and transcription-coupled NER (TC-NER), which preferentially repairs damage in actively transcribed genes. Both converge on a common incision machinery involving the ERCC1-XPF endonuclease complex, which executes the critical 5′ incision step.[40,41] Arsenic inhibits NER through multiple mechanisms: it reduces expression of the damage recognition factor XPC, displaces zinc from the zinc finger domains of NER proteins XPA and PARP-1, and suppresses p53-mediated transcriptional activation of NER genes.[34,35,36] The consequence is that cells co-exposed to arsenic and other carcinogens accumulate unrepaired NER substrates at a higher rate, potentiating carcinogenesis. Importantly, prior work characterizing arsenic-NER interactions has been conducted in race-indeterminate bulk cell populations; whether the magnitude or context-dependence of arsenic-induced NER impairment differs by racial background has not been investigated.

Cancer stem cells (CSCs) represent a self-renewing, tumor-initiating subpopulation implicated in breast cancer initiation, therapy resistance, and recurrence.[6,7] In breast tissue-derived cell systems, CSCs are identified by the surface phenotype CD24−/CD49f+/CD44+: loss of the differentiation marker CD24, retention of the mammary progenitor marker CD49f (integrin α6), and co-expression of the adhesion receptor CD44.[6,7] This marker combination isolates a population with enhanced self-renewal capacity, elevated DNA repair programs, and resistance to genotoxic stress - properties that make CSCs an especially relevant cellular compartment for studying the long-term consequences of arsenic exposure. Despite the established importance of CSCs in breast carcinogenesis, the transcriptional biology of CSCs derived from racially distinct breast tissue-derived cell populations, and the impact of arsenic on these differences, has not been characterized.

This study addresses the intersection of these three underexplored areas - racial biology, arsenic-induced DNA repair impairment, and cancer stem cell transcriptomics - using a genome-wide RNA sequencing approach with a specific design to detect accumulated small changes in NER and DNA repair pathway gene expression rather

than relying on individual gene thresholds. We profiled two breast tissue-derived cell lines representing AA (BRL-24) and NHW (BRL-23) racial backgrounds across parental and CD24−/CD49f+/CD44+ sorted cancer stem cell populations, under vehicle and chronic low-dose arsenic trioxide conditions, generating 12 pairwise contrasts. Pathway-level analysis using Gene Set Enrichment Analysis enabled detection of coordinated, sub-threshold changes across NER gene sets and confirmed stem cell population identity through enrichment of published cancer stem cell signatures. The central finding - that arsenic unmasks a selective suppression of NER specifically in AA cancer stem cells that is absent in NHW stem cells and absent in parental cells of either line - identifies the cancer stem cell compartment as the primary locus of racial biological divergence relevant to arsenic-associated breast cancer risk.

MATERIALS AND METHODS

1. Cell Lines and Maintenance

Two patient-derived breast tissue cell lines were used in this study: BRL-23 (NHW) and BRL-24 (AA), derived from non-diseased breast reduction mammoplasty tissue obtained from premenopausal women under institutional review board approval (UPMC IRB 0609002) and characterized in prior work from this laboratory.[1] Karyotyping confirmed normal chromosomal content for both lines. Cells were cultured in MWRI medium[2] composed of DMEM (HyClone, Cat. SH30243.01) supplemented with 10% heat-inactivated fetal bovine serum (HyClone, Cat. SH30084.03), 10% heat-inactivated newborn calf serum (HyClone, Cat. SH30118.04), 1% penicillin/streptomycin (Corning, Cat. 30-002-CI), 1% MEM non-essential amino acids (Corning, Ref. 25-025-CI), 1% beta-mercaptoethanol (Sigma-Aldrich, Cat. M3148), and 1% nucleoside stock (Sigma-Aldrich), filter-sterilized through a 0.45 µm membrane. Cells were maintained in 12.5 cm^2 flasks pre-coated with a 1:1 dilution of Matrigel (Corning) and DMEM at 37°C with 10% CO_2.

2. Arsenic Trioxide Exposure and Experimental Design

The arsenic trioxide (ATO; arsenic(III) oxide, CAS 1327-53-3; Sigma-Aldrich) dose was established through a cell viability assay as described by Alharbi (2025),[3] identifying the LC10 as 0.3 µM, consistent with arsenic concentrations in contaminated groundwater.[3,4] For the transcriptomic experiment, cells were seeded, allowed to attach for 24 hours, then divided into vehicle (PBS) and arsenic-treated groups. A 40 mM ATO stock solution was prepared in PBS, filter-sterilized (0.22 µm), and stored at 4°C protected from light. Working solutions were prepared fresh before each feeding. Arsenic-treated cells received 0.3 µM ATO; vehicle controls received equivalent PBS volume. Both groups were fed every other day plus one hour before harvest for 14 days (approximately two to three population doublings), modeling chronic environmental arsenic exposure.[3,5]

At the end of the 14-day exposure, cells were harvested for flow cytometric isolation of cancer stem cell-enriched populations, producing four biological groups per cell line: Parental Vehicle (PV), Parental Arsenic (PA), Stem Vehicle (SV), and Stem Arsenic (SA). Parental conditions were collected at n=3 biological replicates per group per cell line; stem cell conditions yielded n=1 per group due to sorting yield constraints. The sample inventory is summarized in Table 1.

Table 1. Sample inventory for RNA sequencing analysis.

Cell Line	Condition	Population	n	Comparison IDs
BRL-23 (NHW)	Vehicle	Parental	3	C1, C2, C3, C5, C7
BRL-23 (NHW)	Arsenic (0.3 µM)	Parental	3	C2, C3, C7
BRL-23 (NHW)	Vehicle	Stem (CD24−/CD49f+/CD44+)	1	C5, C9, C11
BRL-23 (NHW)	Arsenic (0.3 µM)	Stem (CD24−/CD49f+/CD44+)	1	C7, C9, C12
BRL-24 (AA)	Vehicle	Parental	3	C1, C2, C4, C6, C8
BRL-24 (AA)	Arsenic (0.3 µM)	Parental	3	C2, C4, C8
BRL-24 (AA)	Vehicle	Stem (CD24−/CD49f+/CD44+)	1	C6, C10, C11
BRL-24 (AA)	Arsenic (0.3 µM)	Stem (CD24−/CD49f+/CD44+)	1	C8, C10, C12

Total: 16 samples. AA = African American; NHW = Non-Hispanic White. Stem populations isolated by CD24−/CD49f+/CD44+ surface expression.

3. Cancer Stem Cell Enrichment by Flow Cytometry

Cancer stem cell-enriched populations were isolated from both vehicle- and arsenic-treated cultures at the end of the 14-day treatment period using a three-marker surface phenotype: CD24−/CD49f+/CD44+. This panel is established for isolating progenitor-enriched populations from breast tissue-derived cell lines, with CD24− indicating loss of differentiation antigen, CD49f+ (integrin α6) marking mammary epithelial progenitors, and CD44+ identifying the cancer stem cell adhesion receptor.[6,7] Cells were harvested by trypsinization, washed, and resuspended in PBS with 2% FBS for antibody staining. The following antibodies were used: anti-human CD24-FITC (BioLegend), anti-human CD49f-PE (BioLegend), and anti-human CD44-APC (BioLegend), each incubated at 4°C for 20 minutes. Viable cells were identified by DAPI exclusion. Fluorescence minus one (FMO) controls were used to define gating boundaries for each marker. Flow cytometric sorting was performed on a BD FACS Aria instrument; CD24−/CD49f+/CD44+ cells were collected directly into RNAprotect Cell Reagent (Qiagen) for immediate RNA stabilization. The triple-gated population consistently represented 2-5% of viable cells across both lines and conditions.

4. Total RNA Extraction and Quality Assessment

Total RNA was extracted from all 16 samples using the miRNeasy Mini Kit (Qiagen, Cat. 217004) with on-column DNase digestion (RNase-Free DNase Set, Qiagen, Cat. 79254). Parental cells were lysed with QIAzol directly in the flask, phase-separated with chloroform, and bound to RNeasy Mini spin columns

following the manufacturer's protocol. RNA was eluted in 35 µL RNase-free water. Concentration and purity were assessed by NanoDrop Lite spectrophotometry (A260/A280 1.9-2.1 for all samples). RNA integrity was assessed using the Agilent TapeStation 4200 (RNA Screen Tape, Cat. 5067-5576); all 16 samples achieved RIN ≥ 9, confirming high-quality input.

5. RNA Library Preparation and Sequencing

Libraries were prepared using the TruSeq Stranded Total RNA Library Prep Gold kit (Illumina, Cat. 20020598) and sequenced on an Illumina NextSeq 2000 at the Center for Collaborative Research Genomics Core Facility, Nova Southeastern University (NextSeq 500 High Output Kit, 300-cycle; 1200M reads flow cell). All samples were sequenced as 150 bp paired-end reads at 41-72 million read pairs per sample.

6. Pre-Alignment Quality Control and Batch Structure Assessment

Pre-alignment QC was performed in Partek Flow (v11.0.24.0226) using FastQC-equivalent metrics.[8] Arsenic-treated parental samples showed average Phred scores of approximately 33 versus approximately 38 for vehicle and stem samples; stem samples showed elevated %N (0.24% vs. 0.01-0.07%), indicating separate library preparation batches. Because batch structure was inseparable from biological treatment groups, standard ComBat[9] correction could not be applied without removing biological signal; confounding was instead addressed through the full covariate structure of the DESeq2 model (Section 9).

7. Read Alignment and Gene Quantification

Reads were aligned to GRCh38 (hg38) using STAR[10] within Partek Flow against Ensembl release 105 annotation in paired-end mode, a choice supported by systematic benchmarking of spliced alignment programs,[11] achieving 90-92% alignment rates. Gene-level counts were generated using the quantify-to-annotation-model function (equivalent to featureCounts[12]), requiring strict paired-end compatibility, annotated intron boundary matching for junction reads, and a minimum 75% read overlap threshold. A summed count filter of ≥10 reads across all 16 samples yielded a final matrix of 16,682 genes × 16 samples.

8. Normalization and the DESeq2 Workflow

Raw (unnormalized) gene counts served as input to the DESeq2 analysis pipeline within Partek Flow, as required by DESeq2, which performs its own size factor normalization internally. DESeq2 applies the median-of-ratios method[13] as part of its internal workflow: each gene's pseudo-reference count is the geometric mean across all samples; each sample's size factor is the median of the ratios of its counts to these pseudo-references; counts are divided by size factors before statistical modeling. This corrects for both library size and library composition - a distinction critical for arsenic experiments, where global transcriptional suppression by arsenic could bias simpler approaches. Counts Per Million (CPM) normalization, used in prior work from this laboratory (Alharbi, 2025[3]) for simpler two-group designs, corrects only for sequencing depth by dividing counts by total reads. Because CPM uses the total count sum as its denominator, a few highly expressed genes that dominate the library can inflate the apparent expression of all other genes - a composition bias especially problematic when arsenic globally suppresses transcription. DESeq2-normalized counts

produced by this internal workflow were used for all downstream visualization including principal component analysis. Successful normalization was confirmed by box plot inspection showing identical median heights and interquartile ranges across all 16 samples.

9. Differential Expression Analysis: DESeq2 and the Choice Over CPM + Log2FC

Differential expression was performed using DESeq2 version 1.38[13] within Partek Flow. The DESeq2 + GSEA analytical pipeline was designed around three specific goals: (1) to detect small but consistent accumulations of expression change across NER and DNA repair pathway genes that individually fall below significance thresholds but collectively represent a coordinated biological response to arsenic; (2) to validate that sorted stem cell populations were genuinely transcriptionally distinct from parental cells; and (3) to identify racial differences between BRL-23 and BRL-24 across all conditions and cell types. The choice of DESeq2 over the simpler CPM + Log2FC approach - used in prior work from this laboratory for simpler two-group designs - was made because CPM provides no statistical testing framework, no mechanism for detecting accumulated pathway-level changes, no accommodation for the factorial interaction model, and no ability to borrow variance across genes for n=1 stem cell conditions. Calculating fold change and a p-value directly from CPM-normalized counts using a t-test or ANOVA also violates distributional assumptions, as RNA-seq counts are overdispersed count data, not normally distributed continuous measurements.

DESeq2 models counts using the negative binomial distribution, appropriate for overdispersed count data.[13] Gene-wise dispersion is estimated through empirical Bayes shrinkage, critical for n=1 stem cell conditions where within-group variance cannot be estimated directly; DESeq2 imputes variance from the cross-gene dispersion relationship established in the n=3 parental conditions, enabling principled analysis of all 12 comparisons.[14] Log2 fold change shrinkage was applied using apeglm,[15] stabilizing estimates for lowly expressed genes and preventing noisy estimates from dominating GSEA ranking.

The statistical model included all main effects and pairwise interactions:

~ ethnicity + Treatment + Type + ethnicity*Treatment + ethnicity*Type + Treatment*Type

Reference levels: Vehicle (treatment), Parental (cell type), NHW (ethnicity). The ethnicity-by-treatment interaction directly tests whether arsenic affects gene expression differently in AA versus NHW cells; treatment-by-type tests whether arsenic acts differently in stem versus parental cells; ethnicity-by-type tests whether stemness differs between lines. Fourteen pairwise contrasts were defined (C1-C12 plus three interaction contrasts). Individual gene significance: FDR < 0.05 (Benjamini-Hochberg). Standard t-tests and ANOVA were not used because both assume normally distributed data, a condition violated by RNA-seq count distributions; neither can accommodate the overdispersion of count data nor the n=1 stem cell conditions. The prior publication from this laboratory (Alharbi, 2025[3]) appropriately used CPM + paired t-tests for a simpler two-group design; the present study requires the more sophisticated pipeline to handle the factorial model, detect accumulated pathway-level changes, and analyze n=1 stem cell conditions.

10. Gene Set Enrichment Analysis: Rationale, Goals, and Comparison to Alternative Methods

Pathway-level analysis was performed using GSEA version 4.4.0.[16,17] Three specific analytical goals drove the choice of GSEA over alternative methods. **Goal 1: Detecting accumulated small pathway changes in NER and DNA repair.** DNA repair pathway genes are constitutively expressed housekeeping genes whose

biologically meaningful responses to arsenic are often 1.2- to 1.8-fold changes that may not individually exceed FDR correction thresholds. GSEA detects this biology by evaluating whether NER pathway genes cluster consistently in a specific region of the ranked list, accumulating individual small signals into a significant pathway-level enrichment score. **Goal 2: Confirming stem cell population identity.** GSEA tested whether entire published stem cell gene sets (BenPorath, Pece) were enriched in sorted versus parental cells, providing cross-study validation of population identity. **Goal 3: Identifying racial differences.** Comparison of NES values across 12 comparisons in the NES heatmap identified which pathways differ racially, in which direction, and under which conditions.

Over-Representation Analysis (ORA) vs. GSEA. Partek Flow's built-in gene set analysis uses ORA, which requires a significant gene list and discards all sub-threshold genes. Because DNA repair genes show small consistent changes rather than large individual changes, ORA would discard the signal driving the NER finding and cannot detect it.[16]

Gene Set ANOVA vs. GSEA. Gene Set ANOVA[18] tests whether the average pathway expression differs between groups using an F-test. It requires within-group replicates for variance estimation and therefore cannot be applied to any comparison involving the n=1 stem cell conditions (C5-C12). Gene Set ANOVA also weights all pathway genes equally, diluting signal from the small coherent core of regulated genes. GSEA's weighted enrichment statistic weights each gene by its rank value, emphasizing the most significantly regulated genes. Gene Set ANOVA is most appropriate for detecting broad shifts in average expression level in well-powered balanced designs; in this study it was reserved as a potential confirmatory approach for the parental comparisons (C1-C4) where all groups have n=3, but was not the primary method for any comparison.

11. GSEA Implementation and Statistical Metrics

GSEAPreranked was used for all analyses. Settings: No_Collapse (HGNC symbols match MSigDB format), 1000 permutations, weighted enrichment statistic, minimum gene set size 15, maximum 500, fixed seed 149. Total: 48 runs (12 comparisons × 4 databases). Databases: **Hallmark** (50 curated gene sets, minimal overlap[22]); **KEGG legacy** (mechanistic pathway maps with separate NER, BER, MMR, HR gene sets[23]); **Reactome** (hierarchical pathway descriptions enabling identification of specific affected steps[24]); **CGP** (empirically derived signatures including KAN_RESPONSE_TO_ARSENIC_TRIOXIDE and BenPorath/Pece stem cell signatures for population validation.[19,20])

12. Ranked Gene Lists and the Rank Metric

For each of the 12 comparisons, a ranked gene list was constructed as:

$$\text{Rank Metric} = -\log_{10}(\text{p-value}) \times \text{sign}(\text{shrunken log2 fold change})$$

The $-\log_{10}$(p-value) component places the most statistically significant genes at the list extremes. The sign(shrunken L2FC) component encodes direction. Using the apeglm-shrunken fold change prevents lowly expressed genes with unreliable large fold changes from dominating extreme rank positions. Genes with p-values of zero were assigned ±300. This metric integrates both statistical significance and directionality from the DESeq2 model and is the established approach for GSEAPreranked with pre-computed statistics from an external factorial model.[25] Lists were exported as two-column tab-delimited .rnk files with Unix line endings.

13. Significance Criteria and Statistical Metrics

Three GSEA metrics are reported. The **Enrichment Score (ES) and Normalized Enrichment Score (NES)**: GSEA first calculates a raw ES by walking down the ranked gene list from most upregulated to most downregulated, tracking a running sum that increments when a pathway gene is encountered and decrements for non-pathway genes; the peak deviation of this running sum from zero is the ES. The NES is calculated by dividing the observed ES by the mean ES from 1000 permutations, normalizing for gene set size so NES values are comparable across pathways and across all 12 comparisons. Positive NES: pathway genes enriched at top of ranked list (higher in numerator group). Negative NES: enriched at bottom (higher in denominator). Minimum enrichment criterion: |NES| > 1.5. The **Nominal p-value** is the proportion of permuted enrichment scores exceeding the observed score for a single pathway; useful for ranking individual pathway assessments but not the primary significance criterion. The **FDR q-value** is the primary significance metric: the expected proportion of false positives among all gene sets achieving a given NES or better, calculated against the permutation-derived null distribution.[16] The standard GSEA threshold of FDR < 0.25 was applied, justified because: (1) it was validated in the original GSEA publication; (2) gene sets are not independent, reducing the effective number of independent tests; and (3) the permutation-based scoring is itself conservative.[16] Primary significant findings required |NES| > 1.5 AND FDR < 0.25.

The **leading edge subset** comprises the genes before the enrichment score peak - those most responsible for the enrichment signal. Three statistics characterize it: **tags%** (percentage of pathway genes in the leading edge), **list%** (position of the enrichment peak as a proportion of the ranked list), and **signal%** (an integrated signal strength calculated as tags% × (1 − list%)). Leading edge analysis was performed through the GSEA Leading Edge module to identify genes shared between or unique to leading edges from different comparisons.

14. Gene Annotation Recovery

Of 16,682 genes, 1,806 carried the placeholder "---" in Partek Flow output due to absent HGNC symbols in Ensembl release 105. An INDEX MATCH lookup against a BioMart[26] annotation table recovered symbols for a subset; unresolved genes retained their Ensembl identifier. All 16,682 genes were retained in the ranked list regardless of annotation status.

15. Data Availability and Reproducibility

All GSEA analyses used a fixed random seed (149) to ensure reproducibility. Raw FASTQ files and processed count matrices will be deposited in NCBI GEO upon manuscript acceptance. Complete DESeq2 results (all 14 contrasts, 16,682 genes) and GSEA pathway tables (all 48 runs) will be provided as supplementary data.

RESULTS

Section 1. Racial transcriptional differences in parental breast tissue-derived cells and their modulation by arsenic (C1–C4)

Widespread transcriptional differences distinguish BRL-23 and BRL-24 at baseline

Differential expression analysis of vehicle-treated parental cells (C1: AA vs. NHW Parental Vehicle) identified **7,355 significantly differentially expressed genes** (FDR < 0.05) from 16,682 expressed genes, representing 44% of the transcriptome. Expression was higher in BRL-24 (AA) for 3,847 genes and in BRL-23 (NHW) for 3,508 genes. GSEA identified **71 significantly enriched pathways** (|NES| > 1.5, FDR < 0.25). The AA line showed higher oxidative phosphorylation (*HALLMARK_OXIDATIVE_PHOSPHORYLATION*: NES = +1.68, FDR = 0.18); NHW cells showed higher inflammatory signaling (*HALLMARK_INFLAMMATORY_RESPONSE*: NES = −1.69, FDR < 0.25; *HALLMARK_TNFA_SIGNALING_VIA_NFKB*: NES = −1.57). BRL-24 expressed higher *ALDH1A3*, *ITGA6*, *XPC*, and *ERCC2*; BRL-23 expressed higher *SNAI2*, *VIM*, *CD44*, *MSH2*, *PARP1*, and *DDB1*. Although individual NER genes were higher in AA cells, *KEGG_NUCLEOTIDE_EXCISION_REPAIR* showed no significant enrichment (NES = +0.98, FDR = 0.72), indicating these individual gene differences do not reflect coordinated NER pathway activity at baseline. The *KAN_RESPONSE_TO_ARSENIC_TRIOXIDE* gene set trended toward higher expression in NHW cells at baseline (C1: NES = −1.54), suggesting BRL-23 cells are constitutively more transcriptionally primed for arsenic response (Figure 1, Figure 2A, Figure 3A,B).

Arsenic dramatically narrows racial transcriptional differences in parental cells

Under arsenic conditions (C2: AA vs. NHW Parental Arsenic), a **68% collapse in differentially expressed genes** was observed, falling from 7,355 to 2,328. More strikingly, **all 57 significantly enriched pathways in C2 were enriched in the NHW direction**. Key genes reversed direction under arsenic: *ALDH1A3*, significantly higher in AA at baseline, shifted toward NHW under arsenic, as did *SNAI2* and *TGFBR1*. Pathways persisting as NHW-enriched included *HALLMARK_INFLAMMATORY_RESPONSE* (NES = −1.77), *HALLMARK_TNFA_SIGNALING_VIA_NFKB* (NES = −1.76), and *HALLMARK_XENOBIOTIC_METABOLISM* (NES = −1.59) (Figure 1, Figure 2B).

BRL-23 and BRL-24 mount divergent transcriptional responses to arsenic

Within-line arsenic responses (C3: NHW Arsenic vs. Vehicle; C4: AA Arsenic vs. Vehicle) showed BRL-23 mounting a substantially larger response: **705 significant pathways in NHW versus 334 in AA** (2.1-fold difference). NHW cells strongly induced inflammatory and stress programs (*HALLMARK_INFLAMMATORY_RESPONSE*: NES = +2.14; *HALLMARK_HYPOXIA*: NES = +2.05) and suppressed EMT (*HALLMARK_EPITHELIAL_MESENCHYMAL_TRANSITION*: NES = −1.70). In AA cells, arsenic induced *XPC* (L2FC = +0.78, FDR < 0.001) with no equivalent in NHW (L2FC = +0.03, FDR = 0.87), suggesting divergent DNA damage response strategies. Both lines showed *ERCC6* downregulation (BRL-23: L2FC = −0.29, FDR = 0.01; BRL-24: L2FC = −0.53, FDR < 0.001) and Hedgehog induction (C3: NES = +2.00; C4: NES = +1.54). The *KAN_RESPONSE_TO_ARSENIC_TRIOXIDE* gene set was significantly enriched in AA under arsenic (C4: NES = −1.63, FDR < 0.25) with leading edge genes including *HMOX1*, *AKR1C1*, *SLC7A11*, and *CXCL8* (Figure 1, Figure 2C,D, Figure 3C).

Section 2. Transcriptional validation of stem cell-enriched populations (C5-C8)

Sorted populations are transcriptionally validated as genuine stem cells

To enrich for cancer stem cell populations, BRL-23 and BRL-24 cells were sorted using the CD24−/CD49f+/CD44+ surface phenotype. Comparisons of sorted versus parental cells under vehicle conditions (C5: NHW Stem vs. Parental Vehicle; C6: AA Stem vs. Parental Vehicle) confirmed population identity through three independent lines of evidence. First, principal component analysis showed PC1 (81% of total variance) completely separating stem from parental samples across both lines and conditions, confirming cell type as the dominant source of transcriptional variation. Second, GSEA identified **900 significant pathways in NHW (C5) and 877 in AA (C6)** stem versus parental comparisons, far exceeding racial differences under any condition. Third, sorted cells showed coordinate enrichment of canonical stemness programs: *HALLMARK_G2M_CHECKPOINT* (C5: NES = +2.76), *HALLMARK_E2F_TARGETS* (NES = +2.75), *BENPORATH_ES_2* (NES = +1.84), and *PECE_MAMMARY_STEM_CELL_UP* (NES = +1.85). Enrichment of the BenPorath and Pece signatures, derived from independently sorted stem cell populations in published studies, provides cross-study validation that the CD24−/CD49f+/CD44+ fraction represents a genuine stem cell transcriptional state (Figure 1, Figure 2E,F).

DNA repair capacity was broadly elevated in sorted stem cells, with significant enrichment of *REACTOME_BASE_EXCISION_REPAIR*, *REACTOME_MISMATCH_REPAIR*, and *KEGG_HOMOLOGOUS_RECOMBINATION* across C5-C8, providing functional validation of the sorted populations independent of surface markers. Individual repair genes *LIG1* (C7: L2FC = +0.99, FDR < 0.001) and *POLD1* (C7: L2FC = +1.42, FDR < 0.001) were elevated in NHW stem cells; *CDK7* was consistently downregulated (C5-C8: L2FC −0.39 to −0.61, FDR < 0.05 in C5 and C8) (Figure 1, Figure 3A).

NER is specifically suppressed in AA stem cells under arsenic — the primary racial disparity finding

Comparisons of stem versus parental cells under arsenic (C7: NHW Stem vs. Parental Arsenic; C8: AA Stem vs. Parental Arsenic) revealed a striking racial divergence in NER. In AA stem cells, *KEGG_NUCLEOTIDE_EXCISION_REPAIR* was significantly suppressed (C8: NES = −2.75, FDR = 0.18), **the single strongest pathway finding in the entire dataset**, confirmed by Reactome (*REACTOME_NUCLEOTIDE_EXCISION_REPAIR*: NES = −2.68, FDR < 0.25). No equivalent suppression was observed in NHW (C7: NES = +0.86, FDR = 0.64), nor in any parental comparison (C1-C4 NES range +0.80 to +1.10, all FDR > 0.25), establishing that the racial NER disparity requires both stem cell identity and arsenic exposure.

Leading edge analysis of C8 identified 16 suppressed genes in two functional groups: **damage recognition and TFIIH loading factors** (*XPC* L2FC = −0.45, FDR = 0.016; *DDB1* L2FC = −0.33, FDR = 0.035; *CETN2* L2FC = −0.47, FDR = 0.014; *RAD23A* L2FC = −0.46, FDR = 0.016; *ERCC3*; *GTF2H1*; *CDK7* L2FC = −0.61, FDR = 0.005) and **gap-filling and ligation machinery** (*POLD2*, *POLE3*, *RFC1*, *RFC2*, *RPA2*, *RPA3*). The Reactome C8 leading edge additionally included eight COP9 signalosome subunits (*COPS3-8*), extending suppression to chromatin remodeling steps preceding damage recognition. NHW stem cells under arsenic (C7) showed the opposite pattern, with *REACTOME_BASE_EXCISION_REPAIR* significantly elevated (NES = −2.43), indicating opposite DNA repair trajectories: NHW stem cells maintain BER while AA stem cells suppress NER (Figure 1, Figure 4A).

Section 3. Arsenic-induced transcriptional responses within stem cell populations (C9-C10)

Within-stem arsenic comparisons (C9: NHW Stem Arsenic vs. Vehicle; C10: AA Stem Arsenic vs. Vehicle; both n=1, pathway-level GSEA reported) identified **858 significant pathways in C9 and 967 in C10** - the highest pathway count in the dataset. Arsenic induced a shared program in both lines: *HALLMARK_OXIDATIVE_PHOSPHORYLATION* (C9: NES = +2.51; C10: NES = +2.67), *HALLMARK_HEDGEHOG_SIGNALING* (C9: NES = +2.15; C10: NES = +2.01), *HALLMARK_NOTCH_SIGNALING* (C9: NES = +1.79; C10: NES = +1.70), *HALLMARK_HYPOXIA* (C9: NES = +2.19; C10: NES = +2.35), and *HALLMARK_MYC_TARGETS_V1* (C9: NES = +2.05; C10: NES = +1.66). The NFE2L2 antioxidant axis was activated in both lines with *HMOX1* and *SLC7A11* among the highest-ranked genes. Critically, *KEGG_NUCLEOTIDE_EXCISION_REPAIR* was not significantly altered within either stem line under arsenic (C9: NES = +1.08, FDR = 0.99; C10: NES = +0.85, FDR = 0.33), establishing that the NER suppression in C8 and C12 reflects a cell-intrinsic NER expression difference between sorted AA stem cells and their parental counterparts rather than arsenic directly suppressing NER. Senescence and apoptosis programs were not significantly activated in sorted stem cells under arsenic from either line (Figure 1, Figure 2I,J).

Section 4. Racial transcriptional differences are amplified in the stem cell compartment (C11-C12)

Stem cell enrichment amplifies rather than homogenizes racial differences

Direct racial comparisons within sorted stem cells (C11: AA vs. NHW Stem Vehicle; C12: AA vs. NHW Stem Arsenic) revealed dramatic amplification of racial transcriptional differences. C11 identified **582 significantly enriched pathways** and C12 identified **503 significant pathways**, compared to only 71 (C1) and 57 (C2) in parental racial comparisons - a roughly tenfold amplification consistent across all four MSigDB databases under both conditions (Figure 1, Figure 2K,L).

NER racial disparity confirmed and shown to require arsenic

In C12, *KEGG_NUCLEOTIDE_EXCISION_REPAIR* was significantly enriched in NHW cells (NES = −1.87, FDR < 0.25; Reactome: NES = −2.16), confirming lower NER in AA stem cells under arsenic. NER was not significantly different without arsenic (C11: NES = +0.63, FDR = 0.59) - it is the **only major pathway absent in C11 and emerging in C12**, establishing arsenic as the specific trigger that unmasks the racial NER disparity. The C12 leading edge identified 17 suppressed AA genes comprising gap-filling and ligation machinery: *PCNA*, *LIG1* (L2FC = −0.74), *POLD1*, *POLD3*, *POLE*, *POLE2*, *RFC3-5*. Cross-comparison of C8 and C12 KEGG NER leading edges identified four genes present in both - ***CCNH***, ***ERCC1***, ***RFC2***, ***RPA3*** - the most reproducible markers of the racial NER disparity. *ERCC1* showed consistent suppression in AA stem cells across C5-C8 (L2FC −0.44 to −0.67, FDR < 0.05), establishing it as the primary candidate for functional validation (Figure 1, Figure 4A).

NHW cells show higher inflammatory signaling, xenobiotic capacity, and arsenic response priming

The NHW inflammatory advantage was preserved and amplified in stem cells: *HALLMARK_INFLAMMATORY_RESPONSE* was significantly enriched in NHW stem cells in C11 (NES = −2.30, FDR < 0.05) and C12 (NES = −1.99, FDR < 0.05), substantially stronger than in parental comparisons (C1: NES = −1.69). Cross-comparison identified **23 core inflammatory genes** shared across C1 and C11 including *ICAM1*, *IL1A*, *IL1R1*, *IL7R*, *IFITM1*, *LYN*, and *TLR3*. Eighteen additional genes were gained in C11

stem cells including *IL6*, *CXCL8*, *CCL2*, *NFKBIA*, and *IRAK2*. NHW stem cells also showed significantly higher xenobiotic metabolism (C11: NES = −1.82) and constitutively higher *KAN_RESPONSE_TO_ARSENIC_TRIOXIDE* expression (C11: NES = −1.69; C12: NES = −1.65).

AA cells maintain higher oxidative phosphorylation

The AA OxPhos advantage was the most stable racial difference in the dataset: *HALLMARK_OXIDATIVE_PHOSPHORYLATION* was significantly enriched in AA cells in C1 (NES = +1.68), C2 (NES = +1.26), C11 (NES = +1.77, FDR < 0.25), and C12 (NES = +1.72, FDR < 0.25), unaffected by arsenic treatment or stem cell enrichment (Figure 1, Figure 3C).

Integrative summary

Across all 12 comparisons, this study identifies a layered racial transcriptional landscape qualitatively transformed in the cancer stem cell compartment. In parental cells, arsenic dramatically narrows racial differences (68% collapse in DEGs), yet residual differences uniformly favor NHW cells. In stem cells, racial differences are amplified tenfold and prove more resistant to arsenic-driven convergence. Three stable themes predominate: higher OxPhos in AA cells, higher inflammatory and xenobiotic signaling in NHW cells, and a constitutively more arsenic-primed transcriptional state in NHW cells. Against this backdrop, a fourth theme emerges specifically at the intersection of stem cell identity and arsenic exposure: **arsenic unmasks a selective suppression of NER in AA stem cells** - affecting both initiation (XPC, DDB1, ERCC3, GTF2H1) and completion (PCNA, LIG1, RFC3-5) - absent under all other conditions. ERCC1 is the most reproducible single-gene marker of this disparity. These findings identify the cancer stem cell compartment as the primary locus of racial biological divergence and provide a transcriptomics-level mechanistic framework linking AA racial background, arsenic exposure, stem cell biology, and impaired genomic maintenance.

DISCUSSION

Breast cancer racial disparities represent one of the most persistent and deadly inequities in cancer outcomes: African American women experience approximately 40% higher breast cancer mortality than non-Hispanic White women, a gap that has widened over recent decades despite advances in treatment.[28,29] Environmental arsenic exposure is an established carcinogen with evidence for racial variation in both environmental levels and tissue-level sensitivity.[29,30] Despite these observations, the cellular and molecular mechanisms by which racial background modifies vulnerability to arsenic-induced genomic damage have remained entirely uncharacterized. The present study addresses this gap through genome-wide transcriptional profiling of two patient-derived breast tissue cell lines across 12 pairwise comparisons spanning parental and CD24−/CD49f+/CD44+ sorted cancer stem cell populations, vehicle and arsenic conditions, and within-line and cross-line contrasts. The central finding - that arsenic unmasks a selective suppression of NER in African American cancer stem cells absent in non-Hispanic White stem cells under identical conditions - provides the first transcriptomics-level mechanistic framework connecting race, arsenic exposure, breast cancer stem cell biology, and impaired genomic maintenance in a cancer-relevant model system.

Arsenic as a co-carcinogen that inhibits NER: placing the current findings in mechanistic context

Arsenic is classified as a Group 1 human carcinogen, yet its carcinogenic mechanisms differ fundamentally from classical mutagens.[31] Rather than causing characteristic point mutations, arsenic acts primarily as a co-carcinogen by amplifying the genotoxic effects of other environmental insults.[32,33] A key mechanism is its capacity to impair NER, the primary pathway removing helix-distorting lesions induced by UV radiation, polycyclic aromatic hydrocarbons, and arsenic-associated oxidative adducts.[32,34] Arsenic inhibits NER through multiple mechanisms: it reduces XPC and DDB2 expression,[34,35] displaces zinc from the zinc finger domains of NER proteins XPA and PARP-1,[36,37] and suppresses p53-mediated transcriptional activation of downstream NER genes.[38] The racial specificity of the NER suppression identified here - specific to the AA cancer stem cell context, not observed in NHW stem cells under identical conditions, not observed in parental cells of either racial background under any condition - was not anticipated by prior work, which studied arsenic-NER interactions in race-indeterminate bulk cell populations. The suppression operates at both initiation and completion stages simultaneously, with COP9 signalosome suppression[39] extending the failure to chromatin remodeling upstream of damage recognition.

ERCC1 as the keystone marker of racial NER disparity

ERCC1, encoding the obligate partner of XPF in the 5′ NER incision endonuclease complex, is functionally indispensable for both GG-NER and TC-NER.[40,41] ERCC1 is catalytically inactive alone, functioning by positioning the ERCC1-XPF complex at the pre-incision bubble through interactions with XPA and RPA.[40,42] Its consistent suppression across all four AA stem cell comparisons (C5-C8: L2FC −0.44 to −0.67, FDR < 0.05) and presence in both the C8 and C12 KEGG NER leading edges makes it the single most reproducible gene-level marker of the racial NER disparity. Clinically, high ERCC1 expression is associated with platinum chemotherapy resistance.[40] The present findings suggest a complementary implication: low ERCC1 in AA cancer stem cells under arsenic may predict elevated susceptibility to arsenic-induced DNA damage accumulation.

Cancer stem cells as the primary locus of racial biological divergence

The tenfold amplification of racial pathway differences in stem cells relative to parental cells (C11: 582 vs. C1: 71; C12: 503 vs. C2: 57) is the most architecturally striking finding of this study. Cancer stem cells defined by the CD24−/CD49f+/CD44+ phenotype represent a well-validated progenitor-enriched population with self-renewal capacity, elevated DNA repair programs, and resistance to genotoxic stress.[6,7] The sorted fractions displayed canonical stem cell transcriptional profiles confirming their biological identity, including BenPorath and Pece breast cancer stem cell signatures,[19,20] elevated BER, MMR, and HR programs,[21] and Notch, Hedgehog, and hypoxia pathway activation. The arsenic dependence of the racial NER disparity - absent in C11, present in C12 - demonstrates that environmental exposure is required to unmask a latent vulnerability in AA stem cells, consistent with the gene-environment interaction model of cancer health disparities.[43]

Expected, unexpected, and novel findings

Several results confirmed predictions based on the existing literature and provided internal study validation. The **AA OxPhos advantage** (higher across all 12 comparisons) is consistent with published evidence of population-level differences in mitochondrial metabolic gene expression. **Hedgehog induction by arsenic** (C3, C4, C9, C10) is consistent with published evidence that arsenic promotes cancer stem cell expansion through GLI-dependent programs[44,45] while acting as a GLI antagonist in bulk cells. **Stem cell elevation of BER, MMR, and HR** relative to parental cells is textbook cancer stem cell biology[21] and served as internal validation that CD24−/CD49f+/CD44+ sorting successfully isolated a genomically protected, repair-competent population. **Acute EMT suppression by arsenic in NHW cells** is well documented in the short-term exposure literature. **NHW constitutive arsenic response priming** (KAN arsenic response genes higher at baseline in C1) explains the consistently larger NHW transcriptional response to arsenic.

Three findings were unexpected and represent novel observations. First, **the 68% collapse in racial differences under arsenic in parental cells** ran counter to the common assumption that a genotoxic stressor would amplify pre-existing racial biological differences. The shared arsenic stress response homogenized both transcriptomes, with all residual differences uniformly favoring NHW cells - arsenic acted as a transcriptional equalizer at the bulk population level while simultaneously creating new disparities in specific cell subpopulations. Second, **the tenfold amplification of racial differences in stem cells** was unexpected. Stem cells might intuitively be expected to be more similar across populations as an earlier, less differentiated cellular state. The opposite is true: the sorting procedure strips away shared bulk transcriptional programs, revealing a deeper layer of racial biology that is more pronounced in the stem compartment. This implies that bulk-population disparity studies may systematically underestimate the magnitude of biologically meaningful racial differences. Third, **the strict three-condition requirement for the racial NER disparity** - requiring African American background AND stem cell identity AND arsenic exposure simultaneously - was not anticipated by prior work, which did not distinguish racial backgrounds or examine stem cells. NER is the only major pathway to show the specific pattern of absence in C11 and emergence in C12, making this biologically unique in the dataset.

Arsenic maintains self-renewal programs in stem cells of both racial backgrounds

The consistent induction of Hedgehog and Notch by arsenic in sorted stem cells (C9, C10) appears paradoxical given that arsenic is known to inhibit GLI1 directly in bulk cancer cell lines.[44,45] In sorted stem cells, Hedgehog (NES = +2.15 in NHW, +2.01 in AA) and Notch (NES = +1.79 and +1.70) were significantly induced. This may

reflect population dynamics: if arsenic selectively eliminates non-stem cells with high Hedgehog activity while sparing stem cells, the within-sorted-population comparison appears as induction. Non-canonical GLI activation[46] may also operate differently in stem cells. The absence of significant apoptosis or senescence in sorted stem cells under arsenic from either line[21] combined with maintained self-renewal creates conditions under which arsenic-exposed AA stem cells - carrying suppressed NER and elevated OxPhos-driven ROS - could accumulate mutations over time without cytotoxic elimination.

Racial differences in inflammatory signaling and their potential significance

The consistently higher NHW inflammatory tone (23-gene core signature amplified in stem cells) is counterintuitive given that chronic inflammation is a known breast cancer driver and is associated with worse outcomes in African American patients.[47] The NHW inflammatory signature - dominated by interferon-stimulated genes (*IFITM1*), pattern recognition receptors (*TLR3*), and lymphocyte activation markers (*IL7R*, *LYN*) - may represent immune surveillance capacity rather than a classically pro-tumorigenic cytokine state. The amplification in NHW stem cells, adding *IL6*, *CCL2*, *CXCL8*, *NFKBIA*, and *IRAK2*, is consistent with the known role of NFκB signaling in cancer stem cell self-renewal and maintenance.[48]

The arsenic convergence paradox and integrated mechanistic model

The 68% collapse of racial differences under arsenic, with all residual differences favoring NHW cells, reflects the dominance of the shared arsenic stress response over racial biology in the bulk population. The *KAN_RESPONSE_TO_ARSENIC_TRIOXIDE* gene set, already enriched in NHW cells at baseline (C1: NES = −1.54), indicates constitutive NHW arsenic transcriptional priming, explaining both the larger NHW arsenic response (705 vs. 334 pathways) and the uniform residual NHW-direction differences. In stem cells, this convergence fails and the NER racial disparity emerges, reinforcing that arsenic-race interactions are fundamentally different in the stem cell compartment.

An integrated mechanistic model: At baseline, AA stem cells have lower NER gene expression relative to their parental counterparts (reflected across C5-C8). Upon arsenic exposure, NHW stem cells maintain or upregulate BER while AA stem cells' already-lower NER expression falls below the threshold for racial significance, producing the arsenic-dependent racial NER divergence. Arsenic-induced zinc finger protein displacement[36,37] and p53 transcriptional suppression[38] may have larger quantitative effects in cells already expressing less NER machinery. AA stem cells' constitutively higher OxPhos activity generates elevated mitochondrial ROS; arsenic generates additional ROS.[31,33] Higher oxidative DNA lesion burden alongside reduced NER capacity creates a mutagenic double-hit specific to AA stem cells under arsenic - a mechanism consistent with observed racial disparities in breast cancer outcomes and therapy resistance.

Limitations and future directions

The most significant limitation is n=1 per condition for sorted stem cell populations (C5-C12), necessitated by sorting yield constraints. GSEA is more robust to n=1 than individual gene statistics because pathway-level enrichment aggregates signals across 15-500 genes simultaneously, but individual gene-level conclusions from stem cell comparisons should be treated as hypothesis-generating. The NER pathway findings are robust because they represent coordinated suppression of an entire biological program (NES = −2.75, 16 KEGG genes

all in the same direction). Future studies with triplicate stem cell replicates would enable a three-way interaction analysis (ethnicity × treatment × cell type) providing the most direct statistical test of the central hypothesis.

Second, this study uses two patient-derived breast tissue cell lines as proxies for NHW and AA racial biology. Validation in primary breast epithelial cells, organoids, and breast tumor specimens from racially characterized patients is required to determine whether the NER and inflammatory differences are characteristics of the broader racial backgrounds or are idiosyncratic to BRL-23 and BRL-24. Third, the study examines transcriptional abundance, not protein function or actual DNA repair capacity. Direct measurement of NER-specific lesion removal kinetics[34] in sorted populations under arsenic is an essential next step. Finally, why AA stem cells have lower baseline NER gene expression than NHW stem cells remains unresolved. Epigenetic differences at NER gene promoters, given arsenic's well-documented effects on DNA methylation[49] and the known regulation of XPC by CpG methylation, are priority candidates for investigation.

Conclusions

This study provides the first transcriptomics-level evidence that African American and non-Hispanic White patient-derived breast tissue cell lines differ fundamentally in their NER pathway response to arsenic within the cancer stem cell compartment. The analytical pipeline - DESeq2 with a full interaction model followed by GSEAPreranked across four MSigDB databases - was specifically designed to detect accumulated small changes in NER and DNA repair pathway gene expression and to validate stem cell population identity through enrichment of canonical signatures. Arsenic unmasks a selective and reproducible NER suppression in AA stem cells affecting both initiation (XPC, DDB1, CETN2, RAD23A, ERCC3, GTF2H1, CDK7, COP9 signalosome) and completion (PCNA, LIG1, RFC3-5, POLD1, POLE) stages, requiring the specific combination of African American cellular biology, CD24−/CD49f+/CD44+ stem cell identity, and arsenic exposure. ERCC1 is the primary single-gene candidate for functional validation. The tenfold amplification of racial differences in the stem compartment, with NER uniquely unmasked by arsenic, identifies the cancer stem cell as the primary locus of racial biological divergence and provides a mechanistic framework directly relevant to racial disparities in arsenic-associated breast cancer risk.

These findings motivate: functional validation of NER capacity in primary AA and NHW breast epithelial cells and organoids under arsenic exposure; ERCC1 protein quantification in breast tumor specimens stratified by racial background; generation of additional stem cell replicates to enable the three-way interaction statistical tests that would provide the most direct test of the central hypothesis; and prospective epidemiological studies examining NER gene expression as a biomarker of arsenic-associated breast cancer risk in diverse populations.

FIGURES

Figure 1. NES Heatmap — Key Pathway Enrichment Across 12 Comparisons

Normalized Enrichment Score (NES) from GSEAPreranked analysis. 42 curated pathways from four MSigDB databases across 12 pairwise comparisons (C1–C12). Color scale: red = positive NES (pathway enriched in numerator group); blue = negative NES (enriched in denominator); white = no enrichment. **Orange border** = statistically significant (|NES| > 1.5 and FDR < 0.25). Hover any cell for exact values.

−3 (neg) 0 +3 (pos) ★★ SIG: |NES|>1.5 and FDR<0.25 H Hallmark K KEGG R Reactome C CGP

Pathway	C1 AA vs NHW Par. Veh	C2 AA vs NHW Par. Ars	C3 NHW Ars vs Veh	C4 AA Ars vs Veh	C5 NHW Stem vs Par Veh	C6 AA Stem vs Par Veh	C7 NHW Stem vs Par Ars	C8 AA Stem vs Par Ars	C9 NHW Stem Ars vs Veh	C10 AA Stem Ars vs Veh	C11 AA vs NHW Stem Veh	C12 AA vs NHW Stem Ars
NER												
Nucleotide Excision Repair K	+0.98	+0.81	+1.10	+0.80	+0.81	+0.79	+0.86	-2.75	+1.08	+0.85	+0.63	-1.87
Nucleotide Excision Repair R	+0.99	+0.89	+1.21	+1.20	+1.01	+0.68	+1.14	-2.68	+1.36	+1.46	+0.72	-2.16
Global Genome Nucleotide Excision Repair Gg Ner R	+1.04	+0.92	+1.25	+1.24	+1.06	+0.78	+1.23	+0.67	+1.39	+1.53	+0.81	+0.62
Transcription Coupled Nucleotide Excision Repair Tc Ner R	+1.05	+0.94	+1.26	+1.25	+1.10	+0.80	+1.25	+0.74	+1.41	+1.56	+0.84	+0.65
Dual Incision In Gg Ner R	+1.15	+1.06	+1.41	+1.38	+1.27	+1.01	+1.55	+0.97	+1.58	+1.78	+1.01	+0.90
Formation Of Incision Complex In Gg Ner R	+1.27	+1.18	+1.56	+1.50	+1.50	+1.18	+1.89	+1.20	+1.73	+2.01	+1.17	+1.14
Formation Of Tc Ner Pre Incision Complex R	+1.15	+1.06	+1.41	+1.38	+1.27	+1.01	+1.55	+0.97	+1.58	+1.78	+1.01	+0.90
Newman Ercc6 Targets Dn C	+1.52	+1.43	+1.86	+1.70	+2.28	+1.77	+2.43	+1.91	+2.10	+2.35	+1.57	+1.61
BER												
Base Excision Repair R	+0.68	+0.60	+0.93	+0.89	-1.71	-1.73	-2.43	-1.67	+1.08	+1.02	-1.65	-1.42
Base Excision Repair Ap Site Formation R	+0.71	+0.63	+0.95	+0.91	-1.82	-1.77	+0.57	-1.71	+1.10	+1.04	-1.70	-1.45
MMR												
Mismatch Repair R	+0.61	-1.61	+0.88	+0.85	-1.60	-1.67	-1.76	-1.60	+1.03	+0.97	-1.56	-1.37
HR/DSB												
Homologous Recombination K	+0.58	-1.27	+0.79	-1.32	-1.50	-1.57	-1.61	-1.41	+0.88	-1.64	-1.25	-1.26
Hdr Through Homologous Recombination Hrr R	-0.98	-0.92	-1.24	-1.14	-1.07	-1.20	-1.19	-1.22	-1.75	-1.40	-1.05	-1.00
Diseases Of Dna Repair R	-1.23	-1.16	-1.79	-1.62	-1.29	-1.43	-1.42	-1.40	+0.84	+0.75	-1.29	-1.18
BROAD REPAIR												
Dna Repair H	+1.13	+0.96	+1.81	+1.40	+1.24	+1.18	+1.08	+1.38	+2.02	+1.59	+1.24	+1.29
EMT/STEMNESS												
Epithelial Mesenchymal Transition H	-1.22	-1.11	+0.99	-1.29	-1.34	-1.29	-1.26	-1.31	-1.48	-2.27	-1.42	-1.31
Foroutan Integrated Tgfb Emt Dn C	-1.67	-1.46	+1.00	-1.84	+0.94	+0.87	+0.87	+1.02	+0.92	+0.95	-1.95	-1.81
Benporath Es 1 C	-1.36	-1.27	-1.99	-1.41	-1.78	-1.65	-1.57	+0.73	-1.75	-1.97	-1.45	-1.42
Benporath Es 2 C	+1.32	+1.26	+1.66	+1.49	+1.84	+1.57	+1.93	+1.69	+1.72	+1.92	+1.41	+1.44
Lim Mammary Stem Cell Up C	-1.33	-1.25	-1.72	-1.36	-1.70	-1.59	-1.51	-2.25	-1.64	-1.79	-1.41	-1.39
Lim Mammary Luminal Progenitor Up C	-1.34	-1.26	-1.81	-1.38	-1.73	-1.61	-1.53	+0.60	-1.69	-1.85	-1.42	-1.41
Pece Mammary Stem Cell Up C	+1.32	+1.26	+1.66	+1.49	+1.85	+1.57	+1.94	+1.69	+1.72	+1.92	+1.41	+1.44
Boquest Stem Cell Up C	-1.42	-1.32	+0.80	-1.48	+0.52	-1.80	-1.68	+0.84	-1.96	+0.70	-1.53	-1.50
CELL CYCLE												
G2M Checkpoint H	-1.06	-0.97	-1.62	-1.10	-1.18	-1.13	-1.14	-1.15	-1.16	-1.07	-1.24	-1.21
E2F Targets H	-1.33	-1.16	+1.02	-1.37	-1.44	-1.33	-1.37	-1.45	-2.09	+0.83	-1.45	-1.36
Mitotic Spindle H	+0.88	-1.42	+1.26	+1.00	-1.72	-1.72	-1.61	-2.47	+1.14	+1.10	+0.97	+0.50
SIGNALING												
Hedgehog Signaling H	+1.28	+1.08	+2.00	+1.54	+1.97	+1.36	+2.02	+1.55	+2.15	+2.01	+1.46	+1.37
Signaling By Hedgehog R	+0.67	+0.59	+0.92	+0.89	-1.69	-1.72	-2.31	-1.64	+1.07	+1.01	-1.64	-1.41
Hedgehog Ligand Biogenesis R	+1.23	+1.13	+1.52	+1.46	+1.38	+1.10	+1.77	+1.11	+1.67	+1.90	+1.10	+1.07
Notch Signaling H	+1.42	+1.14	+2.06	+1.59	+2.40	+1.53	+2.75	+1.59	+2.43	+2.63	+1.54	+1.37
Tgf Beta Signaling H	+0.93	-1.51	+1.27	+1.03	-1.91	-2.27	-1.80	+0.94	+1.30	+1.18	+1.04	+0.91
Wnt Beta Catenin Signaling H	+1.04	+0.77	+1.43	+1.23	+1.02	+0.95	-1.97	+1.13	+1.56	+1.41	+1.10	+1.09
Myc Targets V1 H	+1.14	+0.98	+1.83	+1.40	+1.40	+1.20	+1.14	+1.42	+2.05	+1.66	+1.33	+1.31
Myc Targets V2 H	+0.79	-1.31	+1.21	+0.93	-1.67	-1.58	-1.52	-1.88	+1.10	+1.01	+0.70	-2.05
STRESS/ARSENIC												
Inflammatory Response H	-1.69	-1.31	+1.15	+0.92	-1.63	-1.56	-1.45	-1.66	+1.03	+0.99	-2.30	-1.99
Oxidative Phosphorylation H	+1.68	+1.26	+2.14	+1.83	+2.48	+1.68	+2.76	+1.61	+2.51	+2.67	+1.77	+1.72
Tnfa Signaling Via Nfkb H	-1.57	-1.29	+1.15	+0.86	-1.61	-1.48	-1.45	-1.62	+1.01	+0.96	-2.23	-1.99
Xenobiotic Metabolism H	-1.38	-1.19	+1.06	-1.50	-1.56	-1.36	-1.37	-1.49	+0.84	+0.85	-1.82	-1.42
Hypoxia H	+1.32	+1.10	+2.05	+1.57	+2.31	+1.45	+2.49	+1.58	+2.19	+2.35	+1.51	+1.37
Uv Response Dn H	-1.53	-1.22	+1.12	+0.75	-1.58	-1.47	-1.43	-1.59	+1.01	+0.94	-1.89	-1.53
Nuclear Events Mediated By Nfe2L2 R	-1.18	-1.11	-1.65	-1.50	-1.25	-1.37	-1.37	-1.37	+0.81	+0.67	-1.22	-1.15
Kan Response To Arsenic Trioxide C	-1.54	-1.39	+0.93	-1.63	+0.86	+0.71	+0.73	+0.96	+0.80	+0.87	-1.69	-1.65

Figure 1. NES Heatmap — Key Pathway Enrichment Across 12 Comparisons.

Normalized Enrichment Score (NES) from GSEAPreranked analysis of 42 curated pathways across 12 pairwise comparisons (C1–C12) spanning parental and CD24−/CD49f+/CD44+ sorted stem cell populations under vehicle and arsenic trioxide conditions (0.3 µM, 14 days) in BRL-23 (NHW) and BRL-24 (AA) breast

tissue-derived cell lines. Each cell displays the NES value for a given pathway in a given comparison; red = positive NES (enriched in numerator group); blue = negative NES (enriched in denominator); white = no enrichment. Color scale clamped at ±3. Pathways are grouped by functional category (NER, BER, MMR, HR/DSB, EMT/Stemness, Cell Cycle, Signaling, Stress/Arsenic). Comparison columns are color-coded by type: racial comparisons (C1, C2, C11, C12), NHW arsenic comparisons (C3, C7), AA arsenic comparisons (C4, C8), NHW stem comparisons (C5, C9), AA stem comparisons (C6, C10). **Orange cell border** = statistically significant (|NES| > 1.5 and FDR < 0.25). Gene set database source: H = Hallmark, K = KEGG, R = Reactome, C = CGP. The primary finding — selective suppression of *KEGG_NUCLEOTIDE_EXCISION_REPAIR* in C8 (AA Stem vs. Parental Arsenic; NES = −2.75, FDR = 0.18) — is the most deeply blue bordered cell, with no equivalent suppression in C7 (NHW Stem vs. Parental Arsenic).

Figure 2. Volcano Plots — Differential Expression (All 12 Comparisons)

Shrunken Log_2 Fold Change (DESeq2 apeglm) vs $-log_{10}$(p-value). Axes zoomed to 98th percentile of bulk genes to show the relevant range clearly. NER genes: blue · Stemness/EMT: orange · Arsenic response: green · Background: gray. Dashed red line = FDR 0.05 threshold. Vertical dashes = ±0.58 (1.5-fold). Hover for gene details.

C1 | C2 | C3 | C4 | C5 | C6 | C7 | C8 | C9 | C10 | C11 | C12

FDR<0.05, |L2FC|>0.58: **202 up / 231 dn** NER sig: **7** Stemness sig: **12**

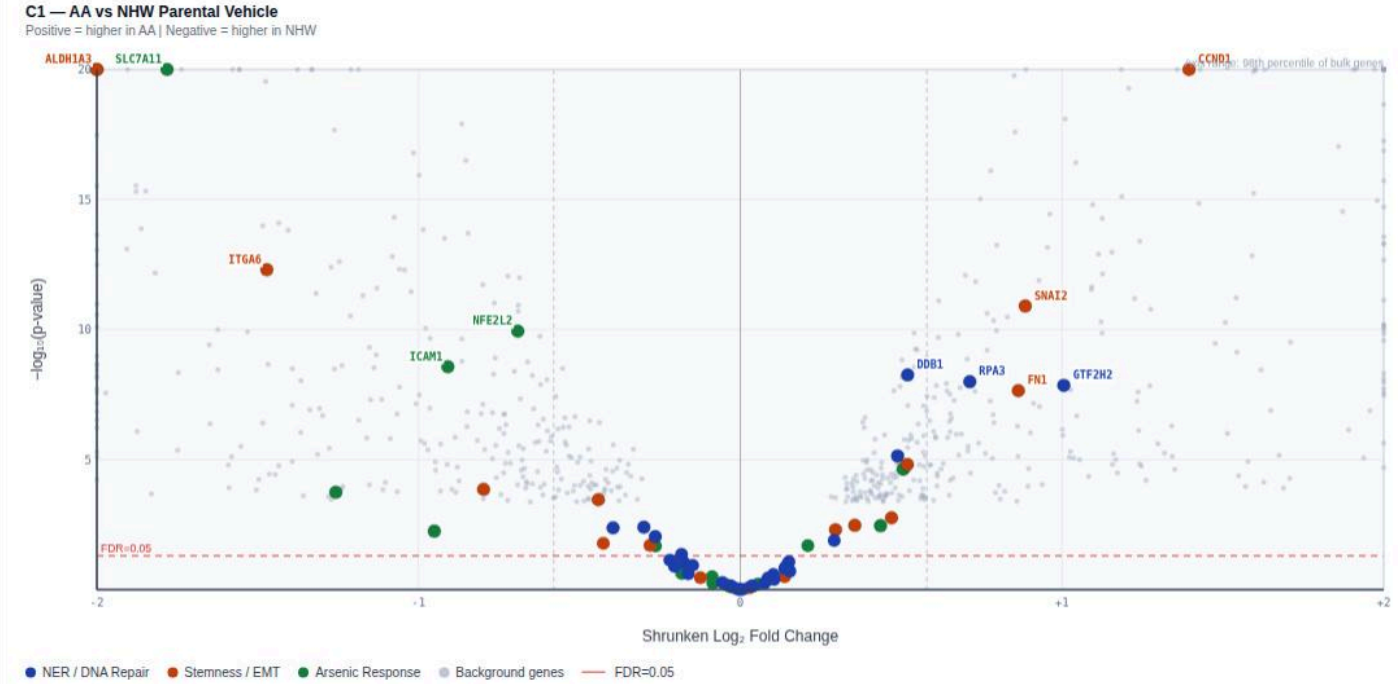

Figure 2. Volcano Plots — Differential Expression (All 12 Comparisons)

Shrunken Log_2 Fold Change (DESeq2 apeglm) vs $-log_{10}$(p-value). Axes zoomed to 98th percentile of bulk genes to show the relevant range clearly. NER genes: blue · Stemness/EMT: orange · Arsenic response: green · Background: gray. Dashed red line = FDR 0.05 threshold. Vertical dashes = ±0.58 (1.5-fold). Hover for gene details.

C1 | C2 | C3 | C4 | C5 | C6 | C7 | C8 | C9 | C10 | C11 | C12

FDR<0.05, |L2FC|>0.58: **181 up / 286 dn** NER sig: **12** Stemness sig: **13**

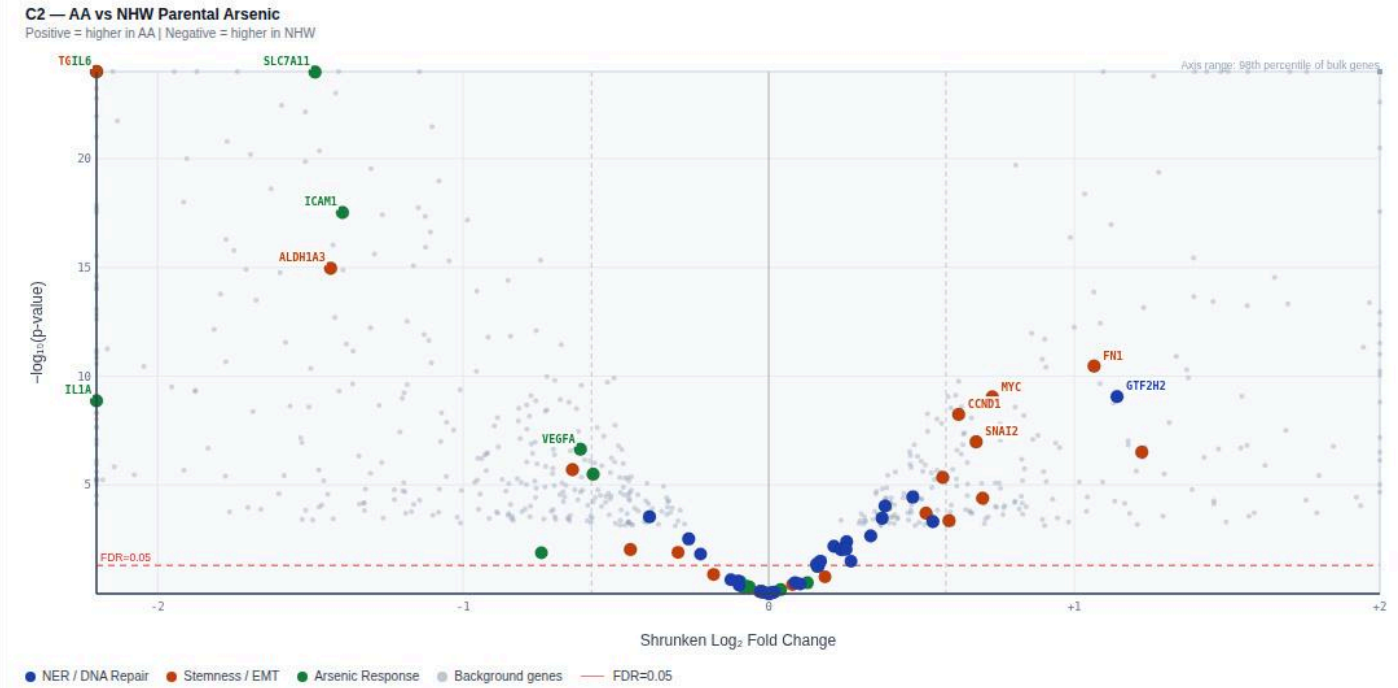

Figure 2. Volcano Plots — Differential Expression (All 12 Comparisons)

Shrunken Log_2 Fold Change (DESeq2 apeglm) vs $-log_{10}$(p-value). Axes zoomed to 98th percentile of bulk genes to show the relevant range clearly. NER genes: blue · Stemness/EMT: orange · Arsenic response: green · Background: gray. Dashed red line = FDR 0.05 threshold. Vertical dashes = ±0.58 (1.5-fold). Hover for gene details.

C1 C2 **C3** C4 C5 C6 C7 C8 C9 C10 C11 C12

FDR<0.05, |L2FC|>0.58: **164 up / 205 dn** NER sig: **7** Stemness sig: **8**

C3 — NHW Arsenic vs Vehicle

Positive = induced by arsenic in NHW | Negative = suppressed by arsenic

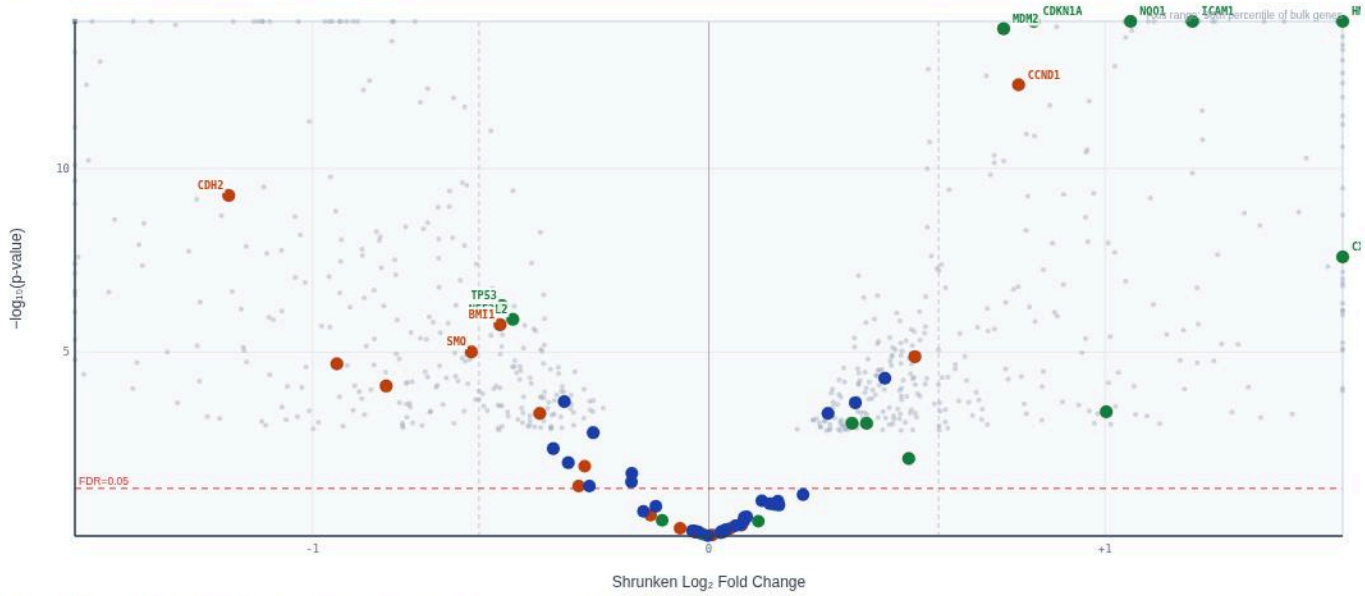

Figure 2. Volcano Plots — Differential Expression (All 12 Comparisons)

Shrunken Log_2 Fold Change (DESeq2 apeglm) vs $-log_{10}$(p-value). Axes zoomed to 98th percentile of bulk genes to show the relevant range clearly. NER genes: blue · Stemness/EMT: orange · Arsenic response: green · Background: gray. Dashed red line = FDR 0.05 threshold. Vertical dashes = ±0.58 (1.5-fold). Hover for gene details.

C1 C2 C3 **C4** C5 C6 C7 C8 C9 C10 C11 C12

FDR<0.05, |L2FC|>0.58: **137 up / 202 dn** NER sig: **9** Stemness sig: **7**

C4 — AA Arsenic vs Vehicle

Positive = induced by arsenic in AA | Negative = suppressed by arsenic

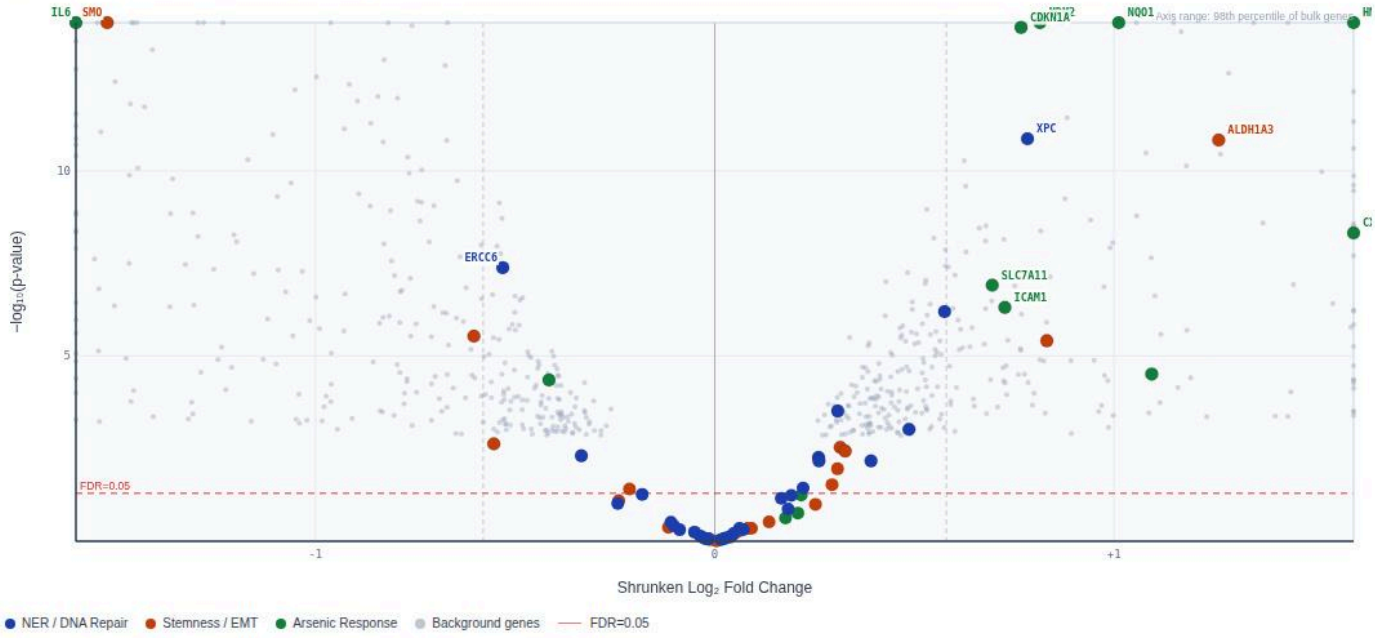

Figure 2. Volcano Plots — Differential Expression (All 12 Comparisons)

Shrunken Log_2 Fold Change (DESeq2 apeglm) vs $-log_{10}$(p-value). Axes zoomed to 98th percentile of bulk genes to show the relevant range clearly. NER genes: blue · Stemness/EMT: orange · Arsenic response: green · Background: gray. Dashed red line = FDR 0.05 threshold. Vertical dashes = ±0.58 (1.5-fold). Hover for gene details.

C1 C2 C3 C4 **C5** C6 C7 C8 C9 C10 C11 C12

FDR<0.05, |L2FC|>0.58: **266 up / 179 dn** NER sig: **9** Stemness sig: **7**

C5 — NHW Stem vs Parental Vehicle

Positive = higher in NHW Stem | Negative = higher in NHW Parental

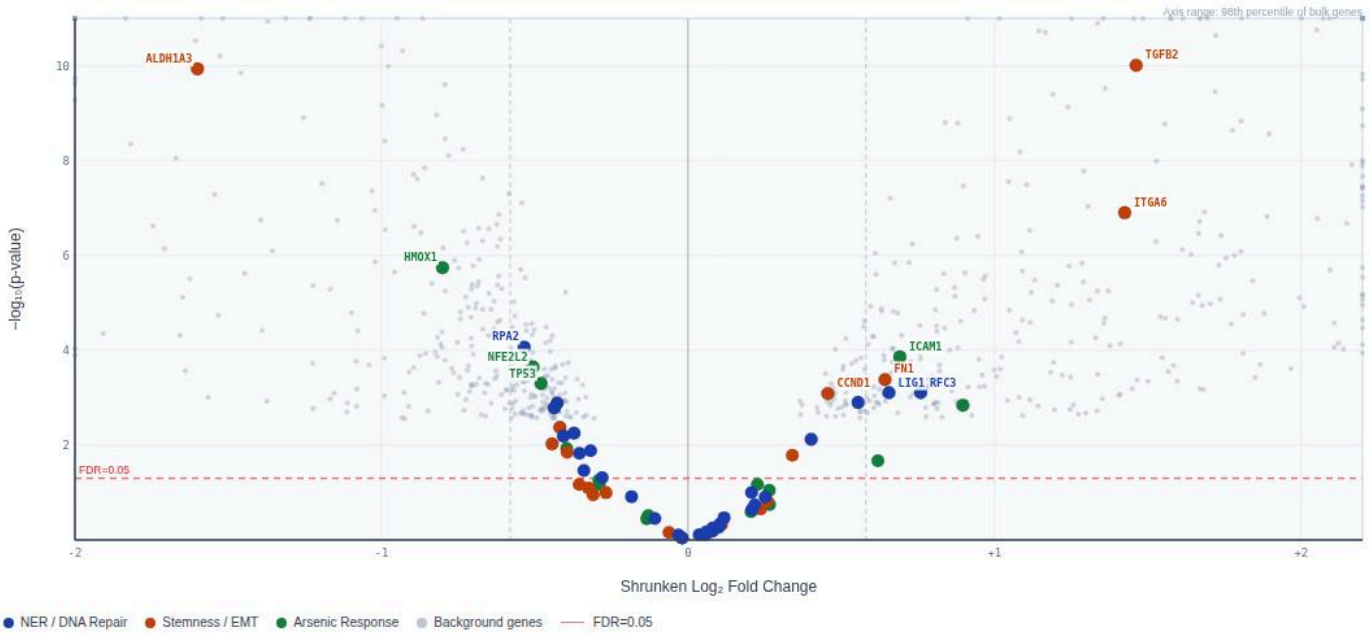

Figure 2. Volcano Plots — Differential Expression (All 12 Comparisons)

Shrunken Log_2 Fold Change (DESeq2 apeglm) vs $-log_{10}$(p-value). Axes zoomed to 98th percentile of bulk genes to show the relevant range clearly. NER genes: blue · Stemness/EMT: orange · Arsenic response: green · Background: gray. Dashed red line = FDR 0.05 threshold. Vertical dashes = ±0.58 (1.5-fold). Hover for gene details.

C1 C2 C3 C4 C5 **C6** C7 C8 C9 C10 C11 C12

FDR<0.05, |L2FC|>0.58: **238 up / 227 dn** NER sig: **6** Stemness sig: **4**

C6 — AA Stem vs Parental Vehicle

Positive = higher in AA Stem | Negative = higher in AA Parental

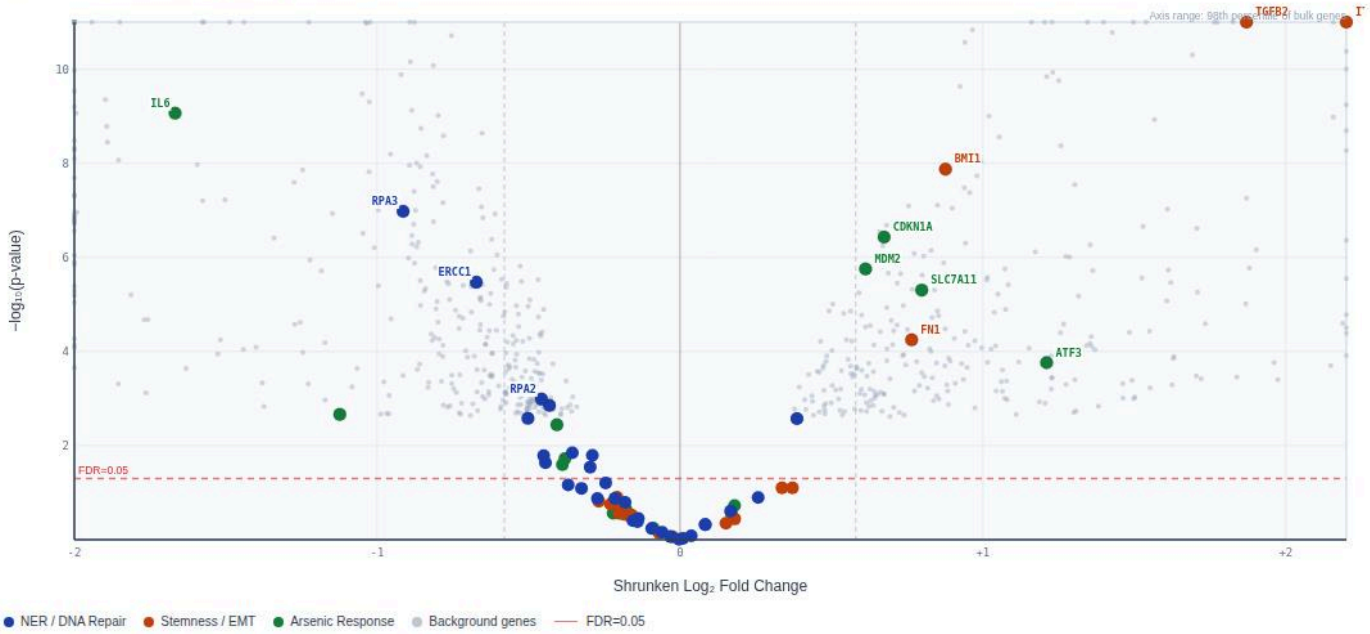

Figure 2. Volcano Plots — Differential Expression (All 12 Comparisons)

Shrunken Log_2 Fold Change (DESeq2 apeglm) vs $-log_{10}$(p-value). Axes zoomed to 98th percentile of bulk genes to show the relevant range clearly. NER genes: blue · Stemness/EMT: orange · Arsenic response: green · Background: gray. Dashed red line = FDR 0.05 threshold. Vertical dashes = ±0.58 (1.5-fold). Hover for gene details.

C1 C2 C3 C4 C5 C6 **C7** C8 C9 C10 C11 C12

FDR<0.05, |L2FC|>0.58: **263 up** / **196 dn** NER sig: **8** Stemness sig: **7**

C7 — NHW Stem vs Parental Arsenic

Positive = higher in NHW Stem | Negative = higher in NHW Parental Ars

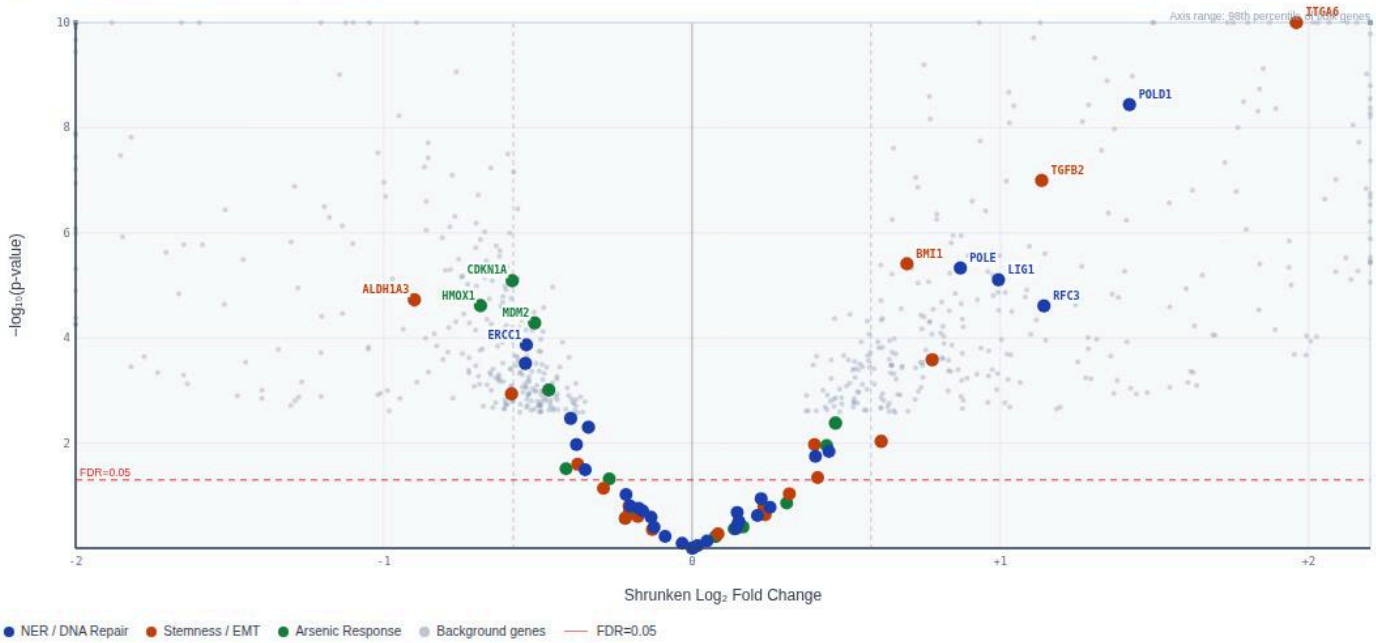

Figure 2. Volcano Plots — Differential Expression (All 12 Comparisons)

Shrunken Log_2 Fold Change (DESeq2 apeglm) vs $-log_{10}$(p-value). Axes zoomed to 98th percentile of bulk genes to show the relevant range clearly. NER genes: blue · Stemness/EMT: orange · Arsenic response: green · Background: gray. Dashed red line = FDR 0.05 threshold. Vertical dashes = ±0.58 (1.5-fold). Hover for gene details.

C1 C2 C3 C4 C5 C6 C7 **C8** C9 C10 C11 C12

FDR<0.05, |L2FC|>0.58: **216 up** / **184 dn** NER sig: **12** Stemness sig: **6**

C8 — AA Stem vs Parental Arsenic

Positive = higher in AA Stem | Negative = higher in AA Parental Ars

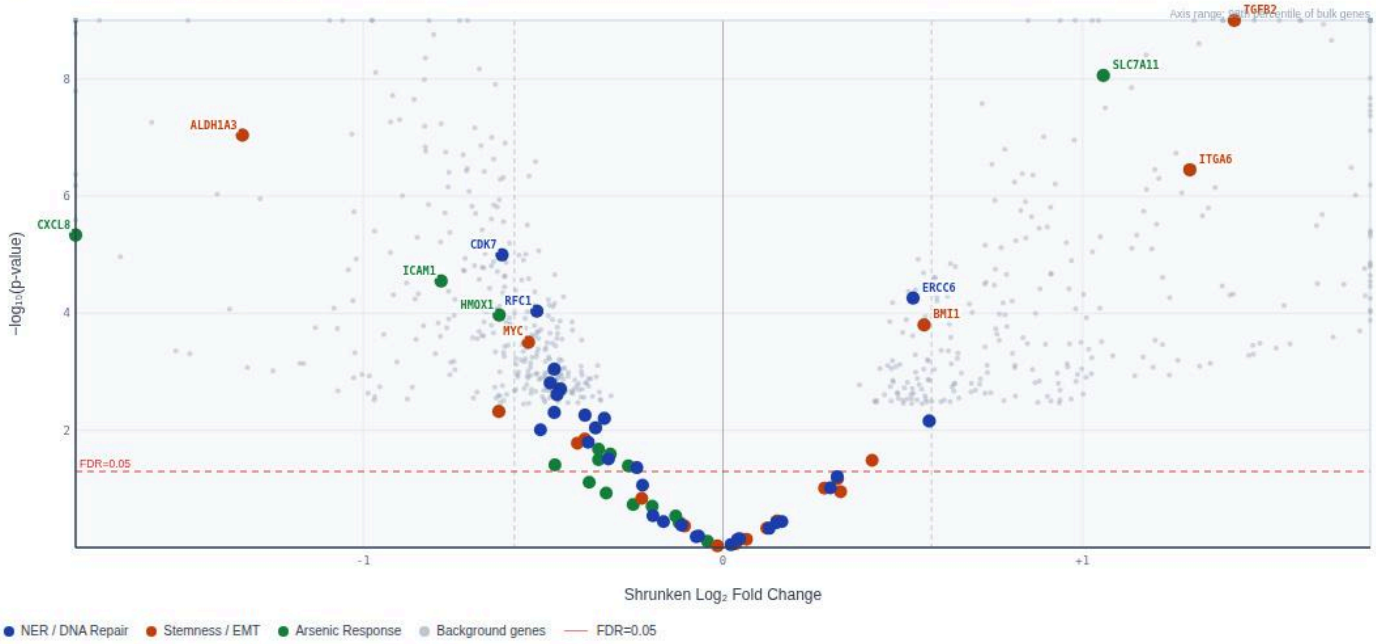

Figure 2. Volcano Plots — Differential Expression (All 12 Comparisons)

Shrunken Log_2 Fold Change (DESeq2 apeglm) vs $-log_{10}$(p-value). Axes zoomed to 98th percentile of bulk genes to show the relevant range clearly. NER genes: blue · Stemness/EMT: orange · Arsenic response: green · Background: gray. Dashed red line = FDR 0.05 threshold. Vertical dashes = ±0.58 (1.5-fold). Hover for gene details.

C1 C2 C3 C4 C5 C6 C7 C8 **C9** C10 C11 C12

FDR<0.05, |L2FC|>0.58: **40 up** / **45 dn** NER sig: **1** Stemness sig: **2**

C9 — NHW Stem Arsenic vs Vehicle

Positive = induced by arsenic in NHW Stem | Negative = suppressed

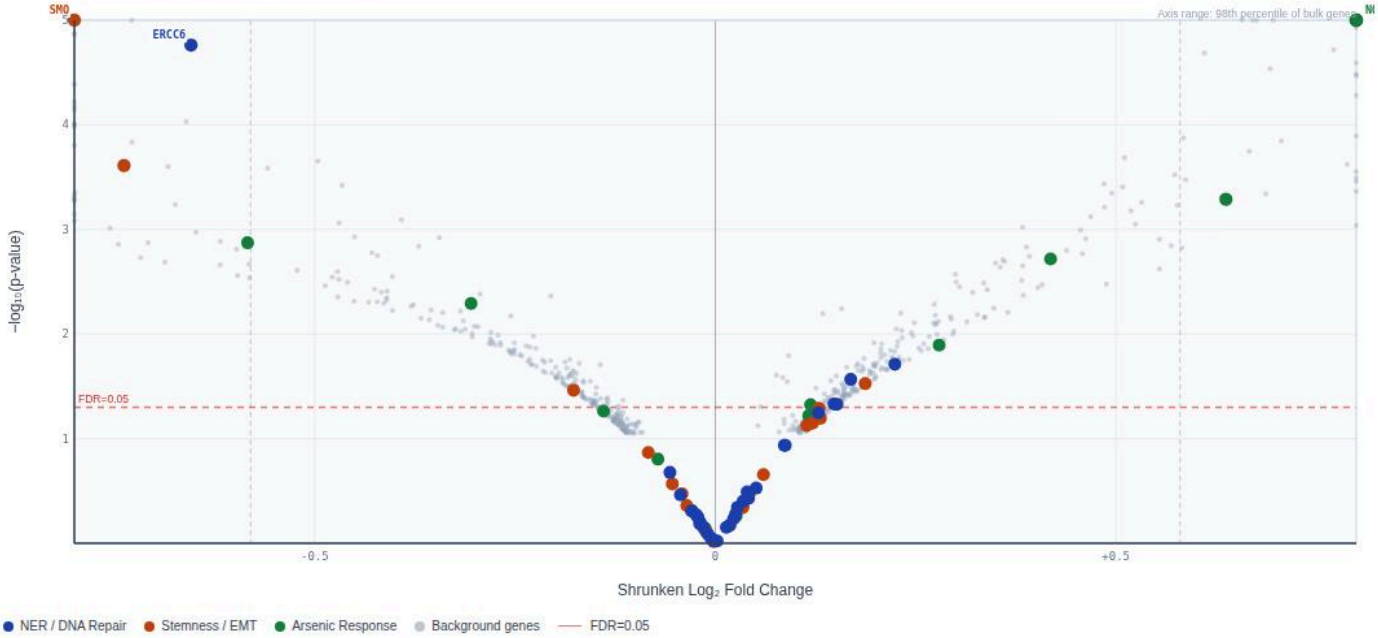

Figure 2. Volcano Plots — Differential Expression (All 12 Comparisons)

Shrunken Log_2 Fold Change (DESeq2 apeglm) vs $-log_{10}$(p-value). Axes zoomed to 98th percentile of bulk genes to show the relevant range clearly. NER genes: blue · Stemness/EMT: orange · Arsenic response: green · Background: gray. Dashed red line = FDR 0.05 threshold. Vertical dashes = ±0.58 (1.5-fold). Hover for gene details.

C1 C2 C3 C4 C5 C6 C7 C8 C9 **C10** C11 C12

FDR<0.05, |L2FC|>0.58: **47 up** / **30 dn** NER sig: **0** Stemness sig: **1**

C10 — AA Stem Arsenic vs Vehicle

Positive = induced by arsenic in AA Stem | Negative = suppressed

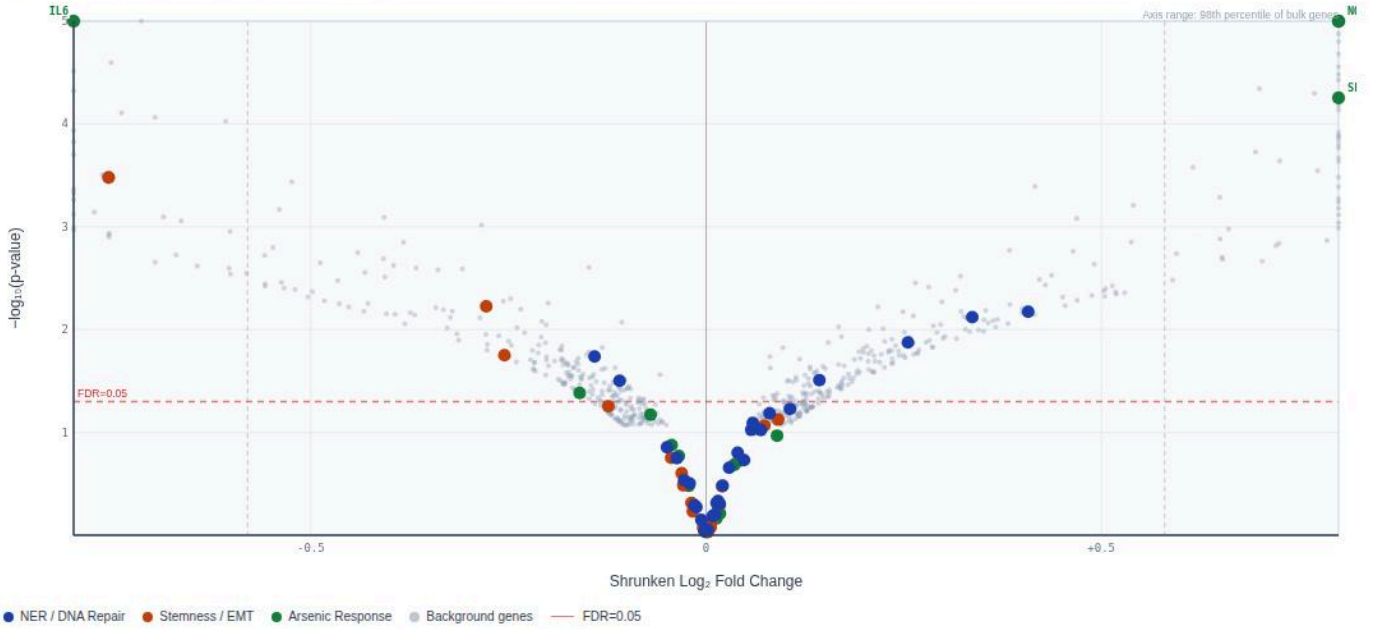

Figure 2. Volcano Plots — Differential Expression (All 12 Comparisons)

Shrunken Log_2 Fold Change (DESeq2 apeglm) vs $-log_{10}$(p-value). Axes zoomed to 98th percentile of bulk genes to show the relevant range clearly. NER genes: blue · Stemness/EMT: orange · Arsenic response: green · Background: gray. Dashed red line = FDR 0.05 threshold. Vertical dashes = ±0.58 (1.5-fold). Hover for gene details.

C1 C2 C3 C4 C5 C6 C7 C8 C9 C10 **C11** C12

FDR<0.05, |L2FC|>0.58: **200 up / 316 dn** NER sig: **8** Stemness sig: **6**

C11 — AA vs NHW Stem Vehicle

Positive = higher in AA Stem | Negative = higher in NHW Stem

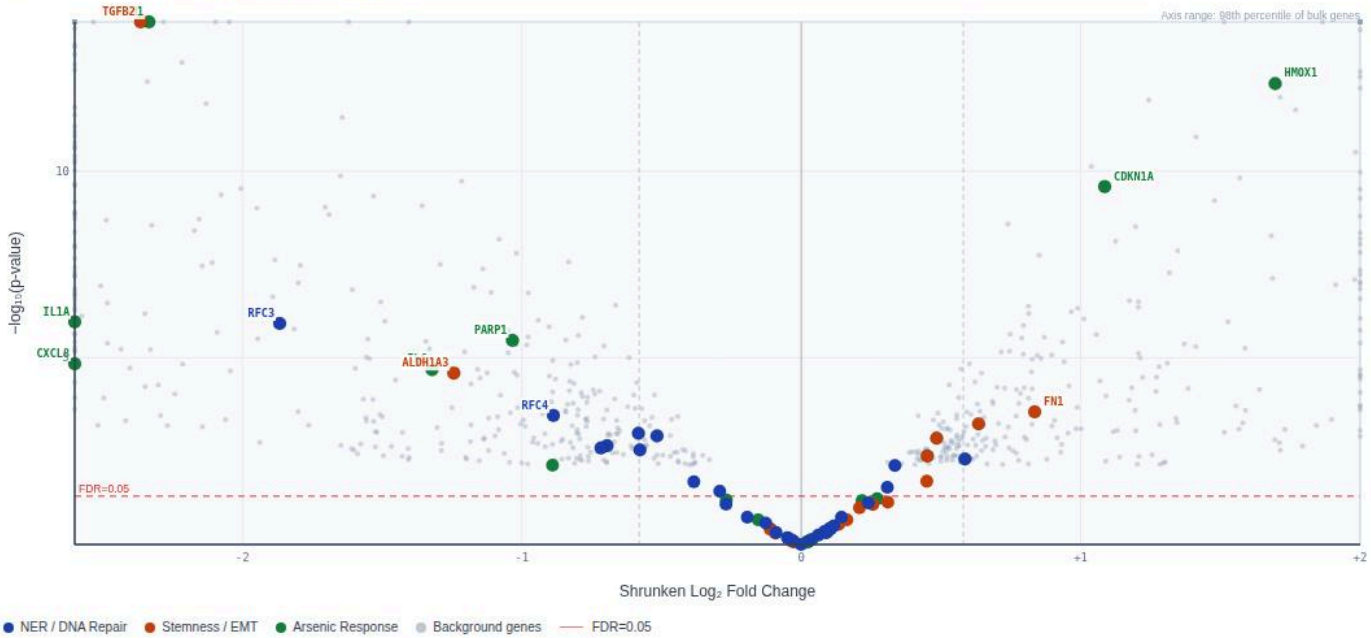

Figure 2. Volcano Plots — Differential Expression (All 12 Comparisons)

Shrunken Log_2 Fold Change (DESeq2 apeglm) vs $-log_{10}$(p-value). Axes zoomed to 98th percentile of bulk genes to show the relevant range clearly. NER genes: blue · Stemness/EMT: orange · Arsenic response: green · Background: gray. Dashed red line = FDR 0.05 threshold. Vertical dashes = ±0.58 (1.5-fold). Hover for gene details.

C1 C2 C3 C4 C5 C6 C7 C8 C9 C10 C11 **C12**

FDR<0.05, |L2FC|>0.58: **198 up / 243 dn** NER sig: **2** Stemness sig: **3**

C12 — AA vs NHW Stem Arsenic

Positive = higher in AA Stem Ars | Negative = higher in NHW Stem Ars

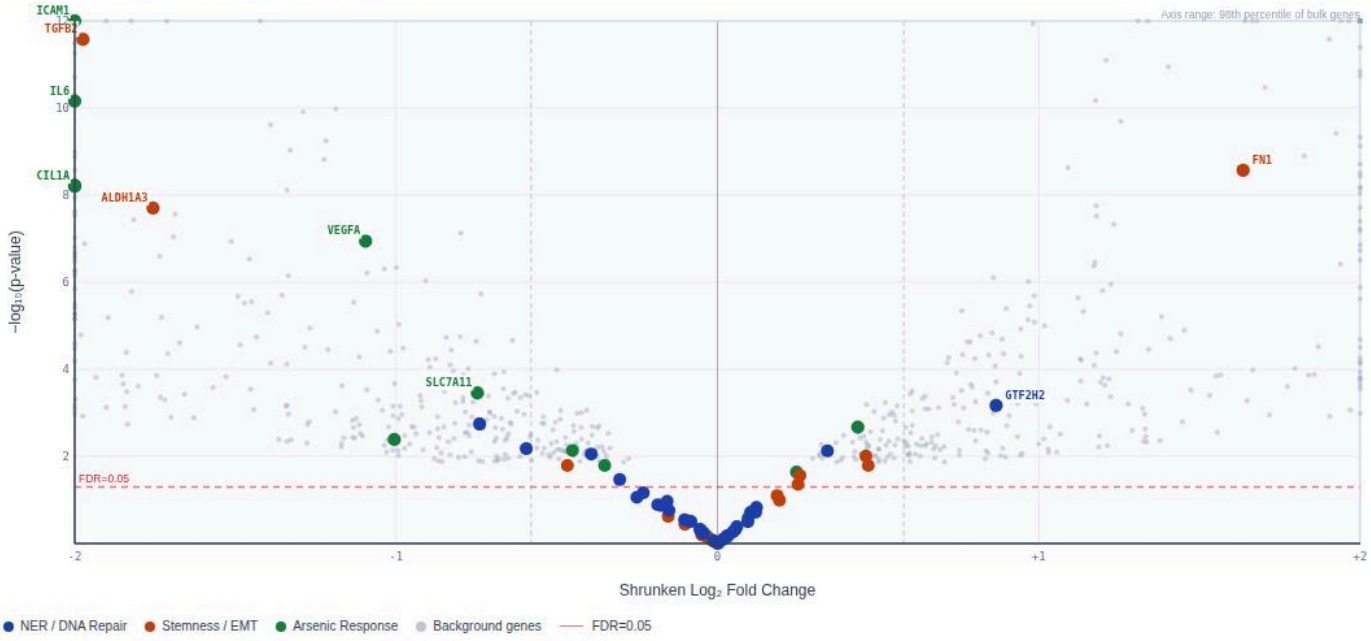

Figure 2. Volcano Plots — Differential Gene Expression Across All 12 Comparisons (Panels A–L).

Volcano plots displaying DESeq2 differential expression results for each pairwise comparison: (A) C1, AA vs. NHW Parental Vehicle; (B) C2, AA vs. NHW Parental Arsenic; (C) C3, NHW Arsenic vs. Vehicle; (D) C4, AA Arsenic vs. Vehicle; (E) C5, NHW Stem vs. Parental Vehicle; (F) C6, AA Stem vs. Parental Vehicle; (G) C7, NHW Stem vs. Parental Arsenic; (H) C8, AA Stem vs. Parental Arsenic; (I) C9, NHW Stem Arsenic vs. Vehicle; (J) C10, AA Stem Arsenic vs. Vehicle; (K) C11, AA vs. NHW Stem Vehicle; (L) C12, AA vs. NHW Stem Arsenic. X-axis: shrunken $\log_2$ fold change (DESeq2 apeglm); y-axis: $-\log_{10}$(p-value). Axes zoomed to the 98th percentile of background genes to display the biologically relevant range clearly. Colored points: **blue** = NER/DNA repair genes; **orange/red** = Stemness/EMT genes; **green** = arsenic response genes; gray = all other genes. Horizontal red dashed line = FDR 0.05. Vertical dashed lines = ±0.58 $\log_2$ FC (1.5-fold). Selected genes are labeled. Gene counts above each panel report significant DEGs (FDR < 0.05, |L2FC| > 0.58). Panels C9 and C10 (n=1 per group) display a compressed axis range reflecting reduced statistical power due to single-replicate stem cell conditions.

Figure 3. Individual Gene Expression — Shrunken Log_2 Fold Change (All 12 Comparisons)

Shrunken Log_2 fold change (DESeq2 apeglm) for key genes across all 12 pairwise comparisons C1–C12. Bars colored by gene category; faded bars = FDR≥0.05 (not significant). Asterisks: * FDR<0.05 ** FDR<0.01 *** FDR<0.001. Bar width scaled within each gene card to its own maximum observed L2FC.

Category: All Genes | NER / DNA Repair | Stemness / EMT | Arsenic Response

NER / DNA Repair (solid = sig) Stemness / EMT Arsenic Response ★ FDR<0.05 ★★ FDR<0.01 ★★★ FDR<0.001

Figure 3. Individual Gene Expression — Shrunken Log_2 Fold Change (All 12 Comparisons)

Shrunken Log_2 fold change (DESeq2 apeglm) for key genes across all 12 pairwise comparisons C1–C12. Bars colored by gene category; faded bars = FDR≥0.05 (not significant). Asterisks: * FDR<0.05 ** FDR<0.01 *** FDR<0.001. Bar width scaled within each gene card to its own maximum observed L2FC.

Category: All Genes | NER / DNA Repair | **Stemness / EMT** | Arsenic Response

NER / DNA Repair (solid = sig) Stemness / EMT Arsenic Response ★ FDR<0.05 ★★ FDR<0.01 ★★★ FDR<0.001

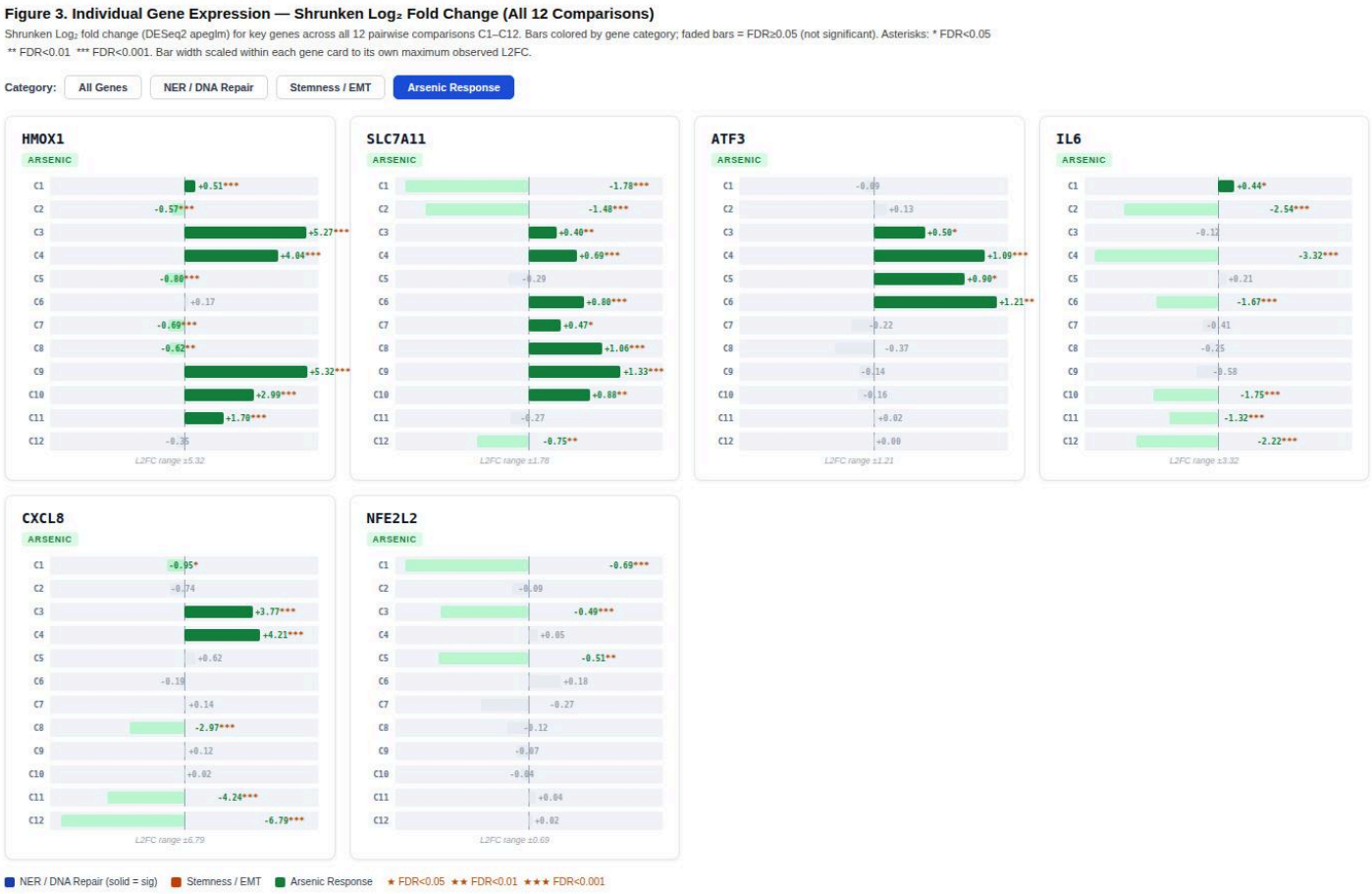

Figure 3. Individual Gene Expression — Shrunken Log$_2$ Fold Change Across All 12 Comparisons (Panels A–C).

Bar charts displaying the shrunken log$_2$ fold change (DESeq2 apeglm) for key individual genes across all 12 pairwise comparisons (C1–C12), organized by gene category. (A) NER/DNA repair genes: ERCC1, XPC, DDB1, DDB2, ERCC3, ERCC6, CDK7, CETN2, RAD23A, POLD1, LIG1, and PCNA. (B) Stemness/EMT genes: SNAI2, VIM, KLF4, BMI1, SMO, ALDH1A3, CD44, and TGFBR1. (C) Arsenic response genes: HMOX1, SLC7A11, ATF3, IL6, CXCL8, and NFE2L2. Bar color indicates gene category (blue = NER/DNA repair; orange/red = Stemness/EMT; green = Arsenic response). Solid bars = FDR < 0.05 (significant); faded bars = FDR ≥ 0.05. Bar width scaled within each gene card to its own maximum observed L2FC. Significance asterisks: * FDR < 0.05; ** FDR < 0.01; *** FDR < 0.001. Positive L2FC = higher expression in the numerator group; negative L2FC = higher expression in the denominator. ERCC1 (panel A) shows consistent suppression specifically in AA stem cell comparisons (C5, C6, C7, C8), establishing it as the most reproducible single-gene marker of the racial NER disparity.

Figure 4. Leading Edge Gene Expression Heatmap

Shrunken Log2 Fold Change (DESeq2 apeglm) for NER pathway leading edge genes across all 12 comparisons (C1-C12). Genes grouped by GSEA leading edge category: shared C8 and C12 (most robust racial signal); C8-only (damage recognition and TFIIH, suppressed in AA Stem vs Parental Arsenic); C12-only (gap-filling and ligation, lower in AA vs NHW Stem Arsenic); additional NER-related. Red = positive L2FC (higher in numerator); blue = negative L2FC; white = no change. Scale clamped at +/-1.5. **Orange border** = FDR < 0.05. **Black dot** = in GSEA leading edge for that comparison. Bottom row = pathway-level NES (** = significant).

-1.5 0 +1.5 | FDR < 0.05 | In GSEA leading edge | **Comparison type:** Racial | NHW Arsenic | AA Arsenic | NHW Stem | AA Stem | Racial Stem

NER Leading Edge Genes | Inflammatory Leading Edge Genes

Gene	C1 AA vs NHW Par. Veh	C2 AA vs NHW Par. Ars	C3 NHW Ars vs Veh	C4 AA Ars vs Veh	C5 NHW Stem vs Par Veh	C6 AA Stem vs Par Veh	C7 NHW Stem vs Par Ars	C8 AA Stem vs Par Ars	C9 NHW Stem Ars vs Veh	C10 AA Stem Ars vs Veh	C11 AA vs NHW Stem Veh	C12 AA vs NHW Stem Ars
SHARED C8 AND C12 -- MOST ROBUST RACIAL NER SIGNAL												
CCNH	-0.18	-0.39	+0.44	+0.22	+0.06	-0.14	-0.38	-0.48 •	-0.00	-0.01	-0.29	-0.39 •
ERCC1	+0.14	-0.22	+0.37	-0.02	-0.44	-0.67	-0.54	-0.47 •	+0.09	+0.02	-0.05	-0.08 •
RFC2	-0.20	+0.10	+0.04	+0.39	+0.05	-0.44	+0.25	-0.51 •	+0.05	+0.05	-0.72	-0.59 •
RPA3	+0.71	+0.33	+0.17	-0.11	+0.40	-0.91	+0.15	-0.37 •	-0.01	+0.08	-0.58	-0.09 •
C8 ONLY -- DAMAGE RECOGNITION AND TFIIH LOADING (SUPPRESSED IN AA STEM VS PARENTAL ARSENIC)												
XPC	-0.30	+0.37	+0.03	+0.78	-0.28	+0.17	-0.20	-0.45 •	+0.03	+0.01	+0.11	+0.06
DDB1	+0.52	+0.17	+0.30	-0.02	-0.18	-0.29	-0.34	-0.33 •	+0.04	-0.01	+0.34	+0.10
ERCC3	-0.00	+0.16	+0.13	+0.31	-0.11	+0.08	-0.13	-0.24 •	+0.03	-0.00	+0.14	+0.03
GTF2H1	+0.04	+0.02	+0.09	+0.07	-0.35	-0.00	-0.54	-0.32 •	-0.02	-0.04	+0.31	+0.12
CDK7	+0.14	+0.09	-0.13	-0.18	-0.43	-0.43	-0.39	-0.61 •	-0.02	-0.11	+0.10	-0.05
CETN2	+0.29	+0.00	-0.04	-0.33	-0.34	-0.50	-0.35	-0.47 •	-0.01	-0.05	+0.09	-0.05
RAD23A	-0.02	+0.47	+0.07	+0.58	-0.41	-0.06	-0.12	-0.46 •	+0.13	+0.02	+0.24	+0.05
RFC1	-0.18	+0.16	-0.20	+0.17	-0.32	-0.18	-0.21	-0.52 •	-0.02	-0.02	-0.03	-0.05
RPA2	-0.15	+0.24	-0.36	+0.05	-0.53	-0.46	-0.03	-0.38 •	+0.03	+0.01	-0.05	-0.04
POLD2	+0.09	+0.38	-0.03	+0.26	-0.37	-0.30	+0.02	-0.35 •	+0.17	+0.03	+0.12	+0.00
POLE3	+0.01	-0.07	-0.01	-0.10	-0.41	-0.48	-0.41	-0.51 •	-0.00	-0.01	-0.04	-0.08
CUL4A	+0.17	-0.01	+0.42	+0.23	-0.07	-0.20	-0.27	-0.23 •	+0.06	+0.03	+0.03	+0.02
C12 ONLY -- GAP-FILLING AND LIGATION (LOWER IN AA VS NHW STEM ARSENIC)												
PCNA	-0.26	-0.10	+0.04	+0.19	+0.08	-0.21	+0.05	-0.11	+0.00	+0.06	-0.52	-0.16 •
LIG1	+0.08	+0.27	+0.24	+0.49	+0.66	-0.16	+0.99	-0.17	+0.15	+0.11	-0.69	-0.74 •
DDB2	+0.04	-0.12	+0.08	-0.10	+0.22	+0.04	+0.00	-0.07	-0.03	-0.03	-0.09	-0.10 •
GTF2H4	+0.01	+0.00	+0.03	+0.03	-0.02	-0.33	+0.14	-0.19	+0.04	+0.02	-0.19	-0.19 •
CUL4B	-0.85	-0.91	-0.17	-0.20	-0.09	-0.08	-0.32	+0.07	-0.12	-0.00	-0.63	-0.16 •
RFC3	-0.16	-0.10	-0.30	-0.24	+0.76	-0.37	+1.14	+0.57	-0.02	+0.41	-1.87	-0.25 •
RFC4	-0.40	-0.02	-0.39	-0.02	-0.02	-0.45	+0.44	+0.13	+0.02	+0.26	-0.89	-0.18 •
RFC5	-0.22	-0.00	-0.03	+0.18	+0.25	-0.14	+0.40	+0.02	+0.02	+0.07	-0.58	-0.23 •
RPA4	+0.06	-0.08	+0.26	+0.07	+0.52	+0.07	+0.33	-0.11	+0.02	-0.02	-0.18	-0.57 •
POLD1	+0.10	+0.54	-0.35	+0.02	+0.21	-0.27	+1.42	+0.30	+0.22	+0.34	-0.27	-0.15 •
POLD3	-0.47	+0.10	-0.51	+0.09	+0.04	-0.37	+0.73	-0.17	+0.04	+0.05	-0.96	-0.79 •
POLE	+0.15	+0.01	+0.18	+0.04	+0.55	-0.09	+0.87	+0.32	+0.15	+0.14	-0.38	-0.30 •
POLE2	-0.29	-0.07	-0.19	-0.01	+0.26	-0.30	+0.75	+0.28	+0.01	+0.21	-1.87	-0.18 •
ADDITIONAL NER-RELATED												
PARP1	-0.26	-0.02	-0.02	+0.22	+0.26	-0.38	+0.15	-0.35	-0.03	+0.04	-1.03	-0.45
PARP2	-0.21	+0.35	-0.14	+0.46	-0.10	-0.01	+0.05	-0.59	+0.01	-0.01	-0.07	-0.14
ERCC6	-0.06	-0.26	-0.29	-0.53	+0.21	+0.39	-0.17	+0.53	-0.65	-0.14	+0.08	+0.34
GSEA NES (pathway)	+1.68	+1.26	+1.86	+1.70	+2.28	+1.77	+2.43	-2.75**	+2.10	+2.35	+1.57	-1.87**

Figure 4. Leading Edge Gene Expression Heatmap

Shrunken Log2 Fold Change (DESeq2 apeglm) for NER pathway leading edge genes across all 12 comparisons (C1-C12). Genes grouped by GSEA leading edge category: shared C8 and C12 (most robust racial signal); C8-only (damage recognition and TFIIH, suppressed in AA Stem vs Parental Arsenic); C12-only (gap-filling and ligation, lower in AA vs NHW Stem Arsenic); additional NER-related. Red = positive L2FC (higher in numerator); blue = negative L2FC; white = no change. Scale clamped at +/-1.5. **Orange border** = FDR < 0.05. **Black dot** = in GSEA leading edge for that comparison. Bottom row = pathway-level NES (** = significant).

-1.5 0 +1.5 | FDR < 0.05 | In GSEA leading edge | **Comparison type:** Racial | NHW Arsenic | AA Arsenic | NHW Stem | AA Stem | Racial Stem

NER Leading Edge Genes | **Inflammatory Leading Edge Genes**

Gene	C1 AA vs NHW Par. Veh	C2 AA vs NHW Par. Ars	C3 NHW Ars vs Veh	C4 AA Ars vs Veh	C5 NHW Stem vs Par Veh	C6 AA Stem vs Par Veh	C7 NHW Stem vs Par Ars	C8 AA Stem vs Par Ars	C9 NHW Stem Ars vs Veh	C10 AA Stem Ars vs Veh	C11 AA vs NHW Stem Veh	C12 AA vs NHW Stem Ars
SHARED C1 AND C11 -- PERSISTENT RACIAL INFLAMMATORY SIGNATURE (PARENTAL AND STEM)												
ICAM1	-0.91 •	-1.39	+1.22	+0.73	+0.69	-0.39	+0.44	-0.78	+0.64	+0.04	-2.33 •	-2.94
IL1A	-1.26 •	-2.45	+1.00	+0.18	+0.06	-1.12	+0.17	-0.47	+0.28	+0.09	-3.42 •	-4.07
IL1R1	-1.98 •	-0.82	-0.01	+1.00	-0.60	+0.22	-1.26	-0.46	-0.12	+0.01	-0.52 •	-0.05
IL7R	-3.19 •	-2.61	+1.38	+1.98	+0.21	+0.18	+0.00	-0.53	+0.97	+0.99	-3.20 •	-3.24
IFITM1	-1.09 •	-0.79	-0.59	-0.31	-1.38	-0.80	-1.64	-1.46	-0.61	-0.71	-0.42 •	-0.43
LYN	-1.47 •	-2.38	+0.91	+0.10	+0.04	-0.20	-0.69	+0.13	-0.00	+0.11	-1.82 •	-0.78
NAMPT	-0.53 •	-1.65	+1.76	+0.66	+0.44	-0.20	-0.14	-0.55	+0.55	+0.03	-1.45 •	-2.13
TLR3	-1.24 •	-0.44	+0.04	+0.32	+0.07	-1.85	-0.07	-0.41	-0.02	+0.45	-3.95 •	-0.79
ADORA2B	-1.37 •	-2.78	+0.12	-0.78	-0.36	-0.93	-0.23	-0.53	+0.05	-0.06	-2.05 •	-3.37
BDKRB1	-0.68 •	-2.62	+2.80	+0.89	+0.53	-0.37	-0.53	-0.17	+0.18	+0.73	-2.47 •	-1.81
CD82	-1.10 •	-2.04	+0.80	+0.14	+0.13	-2.57	-0.11	-2.39	+0.05	+0.02	-4.15 •	-4.44
CXCL6	-2.23 •	-4.61	+3.35	+0.79	+0.56	-0.17	-0.06	-0.74	+0.03	+0.00	-5.06 •	-6.46
C11 ONLY -- AMPLIFIED IN NHW STEM (GAINED IN STEM CELL CONTEXT)												
IL6	+0.44	-2.54	-0.12	-3.32	+0.21	-1.67	-0.41	-0.25	-0.58	-1.75	-1.32 •	-2.22
CXCL8	-0.95	-0.74	+3.77	+4.21	+0.62	-0.19	+0.14	-2.97	+0.12	+0.02	-4.24 •	-6.79
CCL2	-0.49	-1.38	-0.18	-0.48	+0.05	-2.45	-0.34	-2.79	-0.18	-0.13	-3.93 •	-3.92
CCL20	+2.16	-4.40	+3.10	-3.43	+0.01	-5.04	+0.40	-0.83	+3.61	-0.02	-2.70 •	-7.09
NFKBIA	-0.18	-0.08	+0.13	+0.21	+0.27	-0.22	+0.31	-0.32	+0.03	+0.01	-0.89 •	-1.01
IRAK2	-0.09	-0.78	+1.49	+0.68	+0.76	-0.85	+0.44	-0.67	+0.87	+0.52	-2.16 •	-2.30
C1 ONLY -- PRESENT AT BASELINE ONLY (LOST IN STEM CELL CONTEXT)												
HIF1A	-0.35 •	-0.40	-0.20	-0.25	-0.36	-0.08	-0.38	-0.11	-0.07	-0.06	-0.04	-0.06
AHR	-0.59 •	-0.23	-0.01	+0.26	-0.31	+0.05	-0.02	-0.26	+0.08	-0.01	-0.09	-0.39
GSEA NES (pathway)	-1.69**	-1.31	+1.15	+0.92	-1.63**	-1.56**	-1.45	-1.66**	+1.03	+0.99	-2.30**	-1.99**

Figure 4. Leading Edge Gene Expression Heatmaps (Panels A–B).

Shrunken $\log_2$ fold change (DESeq2 apeglm) for GSEA leading edge genes across all 12 comparisons (C1–C12). Red = positive L2FC (higher in numerator); blue = negative L2FC (higher in denominator); white = no change. Scale clamped at ±1.5. **Orange border** = FDR < 0.05. **Black dot** = gene is in the GSEA leading edge for that comparison. Bottom row = pathway-level NES (** = |NES| > 1.5 and FDR < 0.25). Comparison columns are color-coded by type. (A) NER pathway leading edge genes, organized into four groups. Shared C8 and C12: genes present in both the C8 (AA Stem vs. Parental Arsenic) and C12 (AA vs. NHW Stem Arsenic) KEGG NER leading edges, representing the most reproducible racial NER markers (CCNH, ERCC1, RFC2, RPA3). C8-only: damage recognition and TFIIH loading factors suppressed in AA stem cells under arsenic (XPC, DDB1, ERCC3, GTF2H1, CDK7, CETN2, RAD23A, RFC1, RPA2, POLD2, POLE3, CUL4A). C12-only: gap-filling and ligation machinery lower in AA vs. NHW stem cells under arsenic (PCNA, LIG1, DDB2, GTF2H4, CUL4B, RFC3, RFC4, RFC5, RPA4, POLD1, POLD3, POLE, POLE2). Additional NER-related genes shown for context (PARP1, PARP2, ERCC6). (B) Inflammatory pathway leading edge genes, organized into three groups. Shared C1 and C11: genes in the leading edge of both the parental (C1) and stem (C11) racial inflammatory comparisons, representing the persistent cell-intrinsic NHW inflammatory signature (ICAM1, IL1A, IL1R1, IL7R, IFITM1, LYN, NAMPT, TLR3, ADORA2B, BDKRB1, CD82, CXCL6). C11-only: genes gained exclusively in the NHW stem cell inflammatory leading edge, representing the stemness-amplified inflammatory program (IL6, CXCL8, CCL2, CCL20, NFKBIA, IRAK2). C1-only: genes present at baseline only, absent in stem cell context (HIF1A, AHR).

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